

Jonathan Widén & Murat Petrov

Administrative Workload Forecasting

An Exploratory Sequential Mixed Methods Study in the Automotive Industry

Industrial Engineering & Management
Master's Thesis

30 ECTS

Semester: Spring-26
Supervisor: Alexandre Sukhov

Acknowledgements

This thesis was written by Jonathan Widén and Murat Petrov, in close collaboration with the case company. Both students have contributed equally, covering 30 ECTS credits within the Master of Science in Industrial Engineering and Management program.

Firstly, we would like to thank the case company for allowing us to conduct this research during the final semester of our studies. We are also deeply grateful for the guidance and valuable time our supervisor at the case company has provided us with, which has been crucial to the project's success. In addition, we would like to thank the interviewees who have participated in this study and the employees who have provided us with their unwavering support throughout the project.

Secondly, we thank the members of the seminar group for their contribution. In particular, we would like to thank Jana Huck, who acted as a secondary assessor, for the constructive feedback and valuable discussions during the seminars. The seminar group has been a key factor in improving the quality of the thesis.

Lastly, we would like to express our sincere gratitude to our supervisor, Associate Professor Alexandre Sukhov, for his genuine engagement throughout the project. Your thought-provoking and persistent challenging of our ideas has been instrumental in sharpening the quality of this thesis. We are deeply grateful for the time and effort you have dedicated to this thesis.

With gratitude,

Jonathan Widén & Murat Petrov

Karlstad University

28th of May 2026

Abstract

As manufacturing companies operate with greater product variation, administrative work becomes more complex. Consequently, greater product variation makes the upcoming administrative workload harder to predict. Although recent literature has applied machine learning to develop forecast models to predict shop-floor workload, it has not been extended to administrative workload in customization-intensive manufacturing environments. The purpose of this thesis is therefore to investigate how resource planning can be improved by building a better understanding of administrative work practices integrated with the application of machine learning forecasting. This was achieved through an exploratory sequential mixed methods research design, beginning with a qualitative phase followed by a quantitative phase. The qualitative phase mapped out the contextual setting through a sociotechnical systems perspective, while the quantitative phase measured the performance of the developed forecasting models. A single case study was conducted at a global truck manufacturer's customer adaptation department in a Swedish manufacturing facility. Findings from the qualitative phase mapped out the customer adaptation department and informed the selection of input variables for developing a forecasting tool. The findings suggested that administrative workload did not stem solely from order volume; other factors, such as time pressure, rework, and product variation, also amplified it internally. The results from the quantitative phase showed that ensemble-based machine learning methods outperformed baseline, linear, and polynomial regression methods. The study demonstrates that resource planning can be improved by combining contextual understanding with machine learning-based forecasting. This enables organizations to anticipate administrative workload and balance, rather than simply reduce, the trade-offs of complexity in complex, customization-intensive manufacturing environments.

Keywords: *Resource planning, Administrative workload forecasting, Machine learning (ML), Sociotechnical systems (STS), Customization, Mixed methods research (MMR).*

Sammanfattning

Tillverkande företag arbetar idag allt mer med en större produktvariation, vilket medför ett mer komplext administrativt arbete. Följaktligen leder produktvariationen till att det blir svårare att förutsäga den kommande administrativa arbetsbelastningen. Även om forskning på senare tid har använt maskininlärning för att utveckla prognosmodeller för att förutsäga arbetsbelastningen på produktionsgolvet, har detta inte utvidgats till att omfatta administrativ arbetsbelastning i tillverkningsmiljöer med hög grad av kundanpassning. Syftet med denna uppsats är därför att undersöka hur resursplaneringen kan förbättras genom att integrera en bättre förståelse för administrativa arbetsrutiner med tillämpningen av maskininlärningsbaserad prognostisering. Detta uppnåddes genom en explorativ sekventiell forskningsdesign med blandade metoder, som inleddes med en kvalitativ fas, följd av en kvantitativ fas. Den kvalitativa fasen syftade till att kartlägga kontexten, medan den kvantitativa fasen mätte prestandan hos de utvecklade prognosmodellerna. En enskild fallstudie genomfördes vid en global lastbilstillverkares avdelning för kundanpassning i en svensk fabrik. Resultaten från den kvalitativa fasen kartlade avdelningen för kundanpassning och vägledde vilka variabler som skulle användas vid utvecklingen av ett prognosverktyg. Resultaten tyder på att den administrativa arbetsbelastningen inte enbart berodde på ordervolymen utan andra faktorer, såsom tidspress, omarbetningar och produktvariationer, förstärkte den också internt. Resultaten från den kvantitativa fasen visade att ensemblebaserade metoder med maskininlärning presterade bättre än baslinje-, linjära och polynomiska regressionsmetoder. Studien visar att resursplanering kan förbättras genom att kombinera kontextuell förståelse med prognoser baserade på maskininlärning. Detta gör det möjligt för organisationer att förutse administrativ arbetsbelastning och hantera komplexitetens avvägningar, snarare än att enbart försöka reducera den, i komplexa tillverkningsmiljöer med hög grad av kundanpassning.

Nyckelord: *Resursplanering, Prognostisering av administrativ arbetsbelastning, Maskininlärning, Sociotekniska system (STS), Kundanpassning, Blandade metoder.*

Abbreviations

CA	Customer Adaptation
CRISP-DM	Cross-Industry Standard Process for Data Mining
DQA	Deductive Qualitative Analysis
ML	Machine Learning
MMR	Mixed Methods Research
STS	Sociotechnical System Perspective

Table of Contents

1 Introduction **8**
1.1 Problem formulation 9
1.2 Purpose and research questions 9
1.3 Thesis outline 10
2 Theoretical Framework **11**
2.1 Sociotechnical system perspective 11
2.2 Complexity 12
2.3 Forecasting 13
2.4 Data mining 16
2.5 Conceptual framework 16
3 Method **19**
3.1 Research design 19
3.1.1 The DQA framework 20
3.1.2 The CRISP-DM framework 21
3.2 Contextual setting 22
3.3 Ethical considerations 23
3.4 Validity of the research design 23
4 The qualitative phase **26**
4.1 Data collection 26
4.1.1 Choice of participants 27
4.1.2 Unstructured interviews 27
4.1.3 Documentation 29
4.1.4 Participant observations 29
4.2 Data analysis 29
4.3 Findings 30
4.3.1 The CA department mapped out 30
4.3.2 Findings from DQA 31
4.3.3 Business understanding 32
4.3.4 Data understanding 32
4.4 Qualitative findings informing the quantitative phase 33
5 The quantitative phase **35**
5.1 Hypothesis development 35
5.2 Data preparation 35
5.3 Modeling 38
5.4 Evaluation 41
5.5 Results 42
5.5.1 Model performance 42
5.5.2 Residuals 43
6 Discussion **45**
6.1 The generation of administrative workload 45
6.2 The suitability of machine learning for forecasting administrative workload 46

6.3 Improving resource planning through an integrated understanding of administrative work and forecasting 47
6.4 Sustainability in resource planning 49
7 Conclusion 51
7.1 Theoretical contributions 51
7.2 Practical implications 52
7.3 Limitations and future research 53
References 55

1 Introduction

Manufacturing companies are increasingly operating in environments with greater product variation, customized orders, and tightly coupled organizational processes. In business-to-business manufacturing, adaptation to customer needs is largely characterized by products customized to meet specific customer requirements (Hallén et al., 1991). While adaptation through customization is necessary to maintain competitive advantages (Da Silveira et al., 2001; Fogliatto et al., 2012), it also contributes to increased complexity within organizations and increases internal demands for coordination and administrative work. Customization orders require additional administrative activities before manufacturing and assembly compared to standardized products. These activities include planning and coordination, information processing, design preparation, and cross-functional interaction. As customization increases the product complexity, which in turn increases the complexity of processes and systems (ElMaraghy et al., 2012), the administrative workload associated with managing customization orders becomes difficult to predict over time. By complexity, we mean the unpredictability of incoming customization orders (both in volume and in the variation of adaptations) and the internal interactions required to handle them. Altogether, these factors make it difficult to anticipate the resulting administrative workload. Administrative workload refers to the set of non-physical tasks required to process and manage customer orders. The challenge is evident in organizations with high levels of adaptation, where administrative processes must balance product complexity with overhead costs, process productivity, and the increased time required to handle orders (Trattner et al., 2019).

To support decision-making under uncertainty, forecasting has long been used as an important managerial tool (Armstrong, 2001; Hyndman & Athanasopoulos, 2021; Petropoulos et al., 2022). Previous research on forecasting with uncertain conditions in various disciplines shows that there is no one-size-fits-all approach, instead models need to be selected based on contextual factors, such as data availability, forecast horizon, and variable complexity (Kara & Şengüler, 2025; Sepaspour et al., 2025). Within forecasting, time series models allow a target variable to be forecasted through a combination of explanatory variables, which may be both of numeric and categorical nature (Petropoulos et al., 2022). Time-series forecasts outperform judgemental forecasts (forecasts based on expert judgements) when historical data is available (Hyndman & Athanasopoulos, 2021). Furthermore, statistical models are suitable when predictable linear relationships exist. However, where linear relationships between variables are insufficient to capture the complexity of the forecasting context, machine learning (ML) approaches offer a non-linear alternative (Petropoulos et al., 2022).

One such environment in which forecasting administrative workload remains a challenge is customization-intensive manufacturing. This case study focuses on a global truck manufacturer in Sweden, more specifically, a dedicated Customer Adaptation (CA) department that handles all administrative tasks related to customization orders. On average, 57% of all orders contain some degree of CA, creating a highly variable flow of administrative work across three main units: Offering, Design Engineering, and Preparation Engineering. Contrary to standardized production processes, customization-intensive processes introduce a high degree of variability in both volume and complexity (ElMaraghy et al., 2012). Additionally, unpredictability directly affects operational

efficiency (Trattner et al., 2019) and creates problems in handling orders (Kropsu-Vehkapera et al., 2011).

In addition to operational benefits, improved workload forecasting can also contribute to more sustainable organizational practices. Informed resource planning enables organizations to allocate resources more efficiently and reduce operational inefficiencies (De Simone et al., 2025). In contrast, environments characterized by high uncertainty often reduce planning capability and increase reactive decision-making (Lee, 2002). Consequently, reactive decision-making can lead to work intensification for employees, which can negatively affect employees' performance (e.g., problem-solving and decision-making at work) and health (e.g., sleep disturbance) if they are exposed to intensive work for extended periods (Mariappanadar, 2014). In addition, reactive decision-making undermines a firm's ability to operate sustainably because sustainability requires long-term decisions that balance short-term pressures with long-term value creation (Bansal & DesJardine, 2014). By enabling more proactive planning of administrative resources, forecasting tools may therefore support economic, social, and environmental dimensions of sustainability in complex manufacturing contexts.

1.1 Problem formulation

Recent research has demonstrated the feasibility of applying ML to develop forecasting models for shop-floor workload (i.e., the labor hours needed to carry out manufacturing tasks on the shop-floor) (e.g., De Simone et al., 2025). However, this has only been done on a small scale and with limited data sets, which limits its generalizability (De Simone et al., 2025). In healthcare, the potential of forecasting workload using ML is being investigated, given the considerable need for more efficient resource planning (Gatta et al., 2020). Although the CA department has historical data for administrative tasks, it has not been used to generate insights about future planning periods. This represents an untapped resource, as established data mining techniques (Han et al., 2012) and more recent advancements in ML (Sarker, 2021) have demonstrated an ability to extract potentially meaningful patterns from historical datasets in organizational contexts. However, the application of ML-based forecasting to administrative workload in customization-intensive manufacturing environments remains sparse in existing literature. Previous research has examined the applicability of ML-based methods for workload forecasting (De Simone et al., 2025; Gatta et al., 2020), while also emphasizing the need for future research that takes into account the operational environment and the people who work in it when determining future workload using forecasting methods (De Simone et al., 2025). Such approaches have yet to be applied to the specific challenge of forecasting administrative workload in CA environments, where organizational complexity and the variability of customization orders pose particular challenges.

1.2 Purpose and research questions

The purpose of this thesis is to investigate how resource planning can be improved in complex, customization-intensive manufacturing environments by building a better understanding of administrative work practices, integrated with the application of ML-based forecasting models. Specifically, the study examines how ML-based forecasting methods can be applied to predict administrative workload in a department preparing customization orders at a global truck

manufacturer in Sweden. To achieve this, the study maps out how administrative work is generated and how it can be predicted. The purpose is addressed by answering the following research questions.

RQ1: How is administrative workload generated within a complex, customization-intensive manufacturing environment?

RQ2: How does the predictive performance of different forecasting methods compare when applied to administrative workload data in a complex, customization-intensive manufacturing environment?

By answering these research questions, the study generates practical implications and theoretical contributions. Practical implications of this thesis are summarized in an implementation framework with instructions that enable managers responsible for resource planning to integrate forecasting tools into their organizations and make informed decisions based on predictions of future administrative workload. The theoretical contributions are directed to the field of operations management by extending the conceptualization of administrative workload in complex, customization-intensive manufacturing environments. Furthermore, the study extends forecasting research by applying ML-based models to administrative work in complex, customization-intensive manufacturing environments. Additionally, the study contributes to the field of industrial engineering and management with its novel application of the Mixed Methods Research (MMR) design. The novel application can help future researchers and practitioners in the field apply an MMR design to contexts characterized by complex, customization-intensive environments where administrative work is difficult to plan.

1.3 Thesis outline

This study consists of seven chapters. The first chapter introduces the background, situates the studied topic in context, and presents the problem formulation that leads to the research questions. The second chapter establishes a theoretical framework by reviewing the existing literature on sociotechnical systems, complexity, forecasting, and data mining, and concludes with a conceptual framework. The third chapter presents the overarching research design, the study's contextual setting, ethical considerations, and the study's validity. Specifically, the overall research design is an exploratory sequential MMR design, comprising two studies: first, a qualitative phase (chapter four) that, in turn, informs the subsequent quantitative phase (chapter five). The qualitative phase in chapter four consists of data collection, analysis, and findings from the mapping of the CA department and the deductive qualitative analysis. In addition, this chapter also presents the Business Understanding and Data Understanding stages, which inform the subsequent quantitative phase. The quantitative phase in chapter five covers the stages of Data Preparation, Modeling, Evaluation, and the results of the forecasting models. The sixth chapter discusses the findings and the results in relation to the research questions, the literature, and the conceptual framework. The chapter also reflects on how the MMR design strengthens the credibility, validity, and reliability of the study's conclusions. Additionally, sustainability in resource planning is also discussed in this chapter. The seventh and final chapter presents the conclusions, theoretical contributions, practical implications, limitations, and directions for future research.

2 Theoretical Framework

This chapter presents the central concepts covered in this study. The concepts laid the foundation for the context and methods of forecasting, which later converged into a conceptual framework for this study.

2.1 Sociotechnical system perspective

To provide a distinct perspective on the underlying drivers of administrative workload, a Sociotechnical System (STS) perspective is adopted. At its core, STS refers to a system consisting of two interacting, distinct subsystems, a social and a technical subsystem (Bostrom & Heinen, 1977; Kemp et al., 2024). The social subsystem refers to the individual characteristics of the workforce, particularly their professional competencies, attitudes, and values (Kemp et al., 2024). Similarly, the technical subsystem accounts for processes, technical hardware, and tasks needed to convert an input into outputs (Kemp et al., 2024). The outputs of the sociotechnical system represent the work produced by these interactions, encompassing both the quantity and quality of tasks completed within the system (Bostrom & Heinen, 1977).

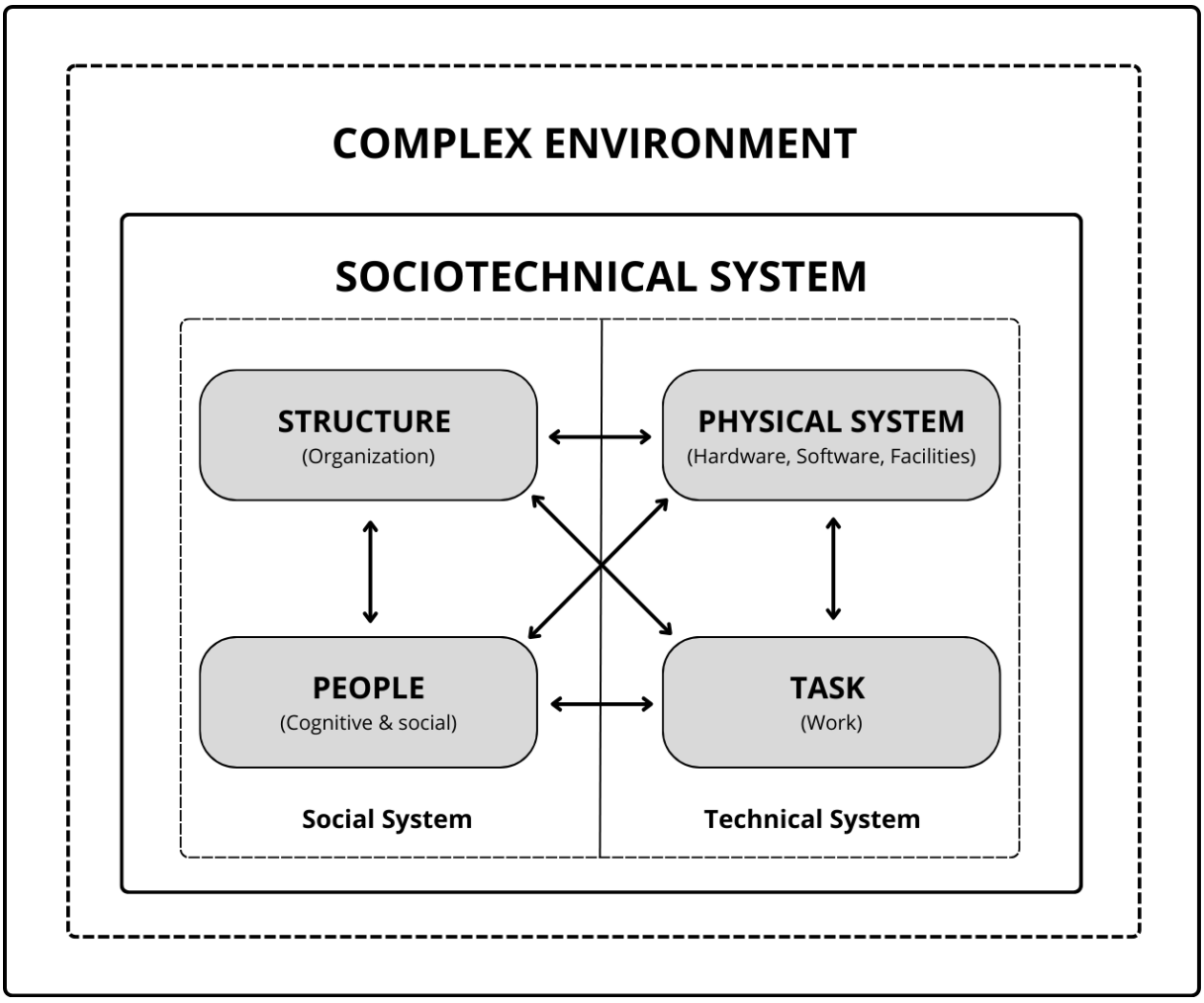


Figure 1: The sociotechnical system. Own figure, adapted from Bostrom and Heinen (1977) and Kemp et al. (2024).

Figure 1 represents the four different elements of an STS. Here, "structure" refers to the foundation and boundaries of a system studied (e.g., a department within a company). "People" refers to individuals or actors within the studied system, and the study explores their emotions and attitudes towards the organization or its daily operations. Furthermore, "physical systems" refers to the technical components, such as hardware and software, that a company relies on for its workflows and communication. Lastly, "task" outlines the subdepartment's actual goals or functions. By studying the task element of an STS, we can identify the resources needed and how tasks are performed.

The elements presented in Figure 1 are not independent, as changes to any element will affect the others. STS outputs result from interactions between the social and technical subsystems, meaning that any change to a system must simultaneously address both subsystems (Bostrom & Heinen, 1977). To understand how these interactions operate, Bostrom and Heinen (1977) distinguish between primary and secondary changes. Primary changes are the intended modifications made to a specific element, such as introducing a new technology. Secondary changes are subsequent shifts in the other subsystems that result directly from the primary change. Modifying how tasks are organized, sequenced, or allocated will alter how people coordinate, communicate, and prioritize their work. Conversely, shifts in how people approach their work will influence how technical systems are used and how future modifications to them are designed (Bostrom & Heinen, 1977).

The STS perspective helps to identify which elements generate administrative work and how they interact. However, the STS perspective does not account for what happens when multiple elements interact simultaneously and repeatedly. These interactions produce non-linear effects in which a single additional input does not result in a proportional increase in output. Understanding this requires looking beyond the system's structure, which is why the study's framework incorporates complexity as a concept.

2.2 Complexity

A complex system is characterized by uncertainty, which makes its outputs not fully predictable (ElMaraghy et al., 2012). Moreover, Anderson (1999) explains that complexity does not arise from the scale of a system, but from the non-linear interactions among its independent parts. This means that a small increase in input can trigger a disproportionate (non-linear) cascade in the output rather than a simple one-to-one increase. Therefore, as the number of interactions increases, the system's complexity increases accordingly (Anderson, 1999). The literature suggests that organizational complexity increases with the number of interactions, which, in this study, refers to interactions among STS elements. However, to understand how such interactions generate complexity in an organizational setting, it is necessary to consider the levels at which complexity can occur. To structure this, ElMaraghy et al. (2012) propose a categorization of complexity scopes according to: (i) part, (ii) product, (iii) system, and (iv) system of systems (e.g., STS), where the complexity in the smaller scope contributes to the complexity in the larger scope.

Trattner et al. (2019) demonstrate, through a systematic literature review, that a complex product requires more resources to handle. Consequently, organizations that handle high levels of customization face compounding complexity across multiple organizational levels simultaneously.

Furthermore, the tangible effects of this compounded complexity are directly prominent in operational performance (Trattner et al., 2019). It has been demonstrated that product complexity has a consistent negative relationship with manufacturing costs, lead time, processing time, and process productivity (Trattner et al., 2019). In the context of order handling, it is further illustrated that product complexity creates problems in operational processes, such as order fulfillment, sales, and delivery (Kropsu-Vehkapera et al., 2011). Moreover, these problems manifest as delays, configuration errors, and difficulties in outlining the company's offerings (Kropsu-Vehkapera et al., 2011).

When the outcomes of a system are difficult to anticipate, forecasting becomes necessary to support resource planning. To select an appropriate forecasting method, it is essential to understand these non-linear interactions, as the models must capture patterns that do not follow proportional relationships between input and output values.

2.3 Forecasting

Forecasting tools can help managers navigate uncertainty and make informed decisions (Armstrong, 2001). Forecasting means using all available information, including historical data and relevant information about upcoming events, to produce the most accurate possible view of what will happen (Hyndman & Athanasopoulos, 2021). However, the choice of forecasting method should be based on the case context (Kara & Şengüler, 2025; Sepaspour et al., 2025). Figure 2 illustrates the selected forecasting approach for the study as a decision tree.

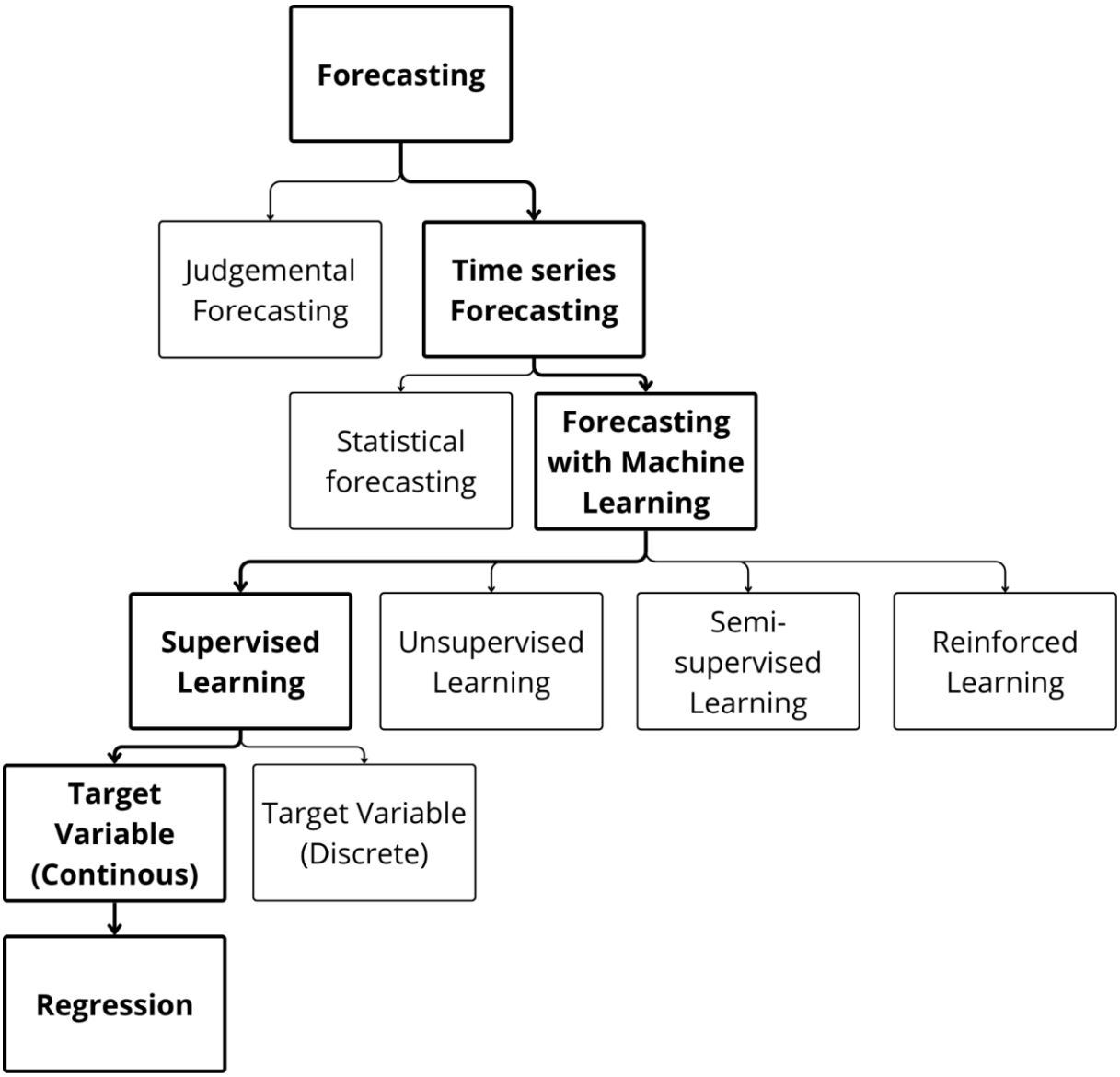


Figure 2: Forecasting approach within the studied context. Adapted from Sarker (2021).

Judgemental forecasts, such as the Delphi method, which relies on a group of experts, can be used when other data are lacking (Hyndman & Athanasopoulos, 2021). However, when historical data is available, and the forecast is time-dependent, as in this study, time-series forecasts are preferable to judgemental forecasts (Hyndman & Athanasopoulos, 2021).

There are misconceptions about whether models belong to the field of statistical or machine learning forecasting (Petropoulos et al., 2022). Models often have their origins in classical statistics but are also used in machine learning forecasting. In classical statistics, the model is predetermined (Petropoulos et al., 2022). In contrast, the process of generating decisions and identifying patterns for a machine without explicit programming is defined as ML (Alam et al., 2025). Machine learning is often defined as a field where the model does not assume a specific mathematical structure in advance, but learns patterns from its input data and makes predictions based on them (Alam et al., 2025; Petropoulos et al., 2022). Therefore, regression models exist within both disciplines, and it is the model's design and application that determine which field it belongs to. Since no predefined

equations can describe the data in the study, ML-based forecasting is chosen to allow the models to identify patterns.

Sarker (2021) proposes two phases for developing a predictive ML-based model (see Figure 3). In the model training phase, ML algorithms are trained on historical data to develop a predictive model. Then, during the model testing phase, new data is used as input to the predictive model to yield predictions.

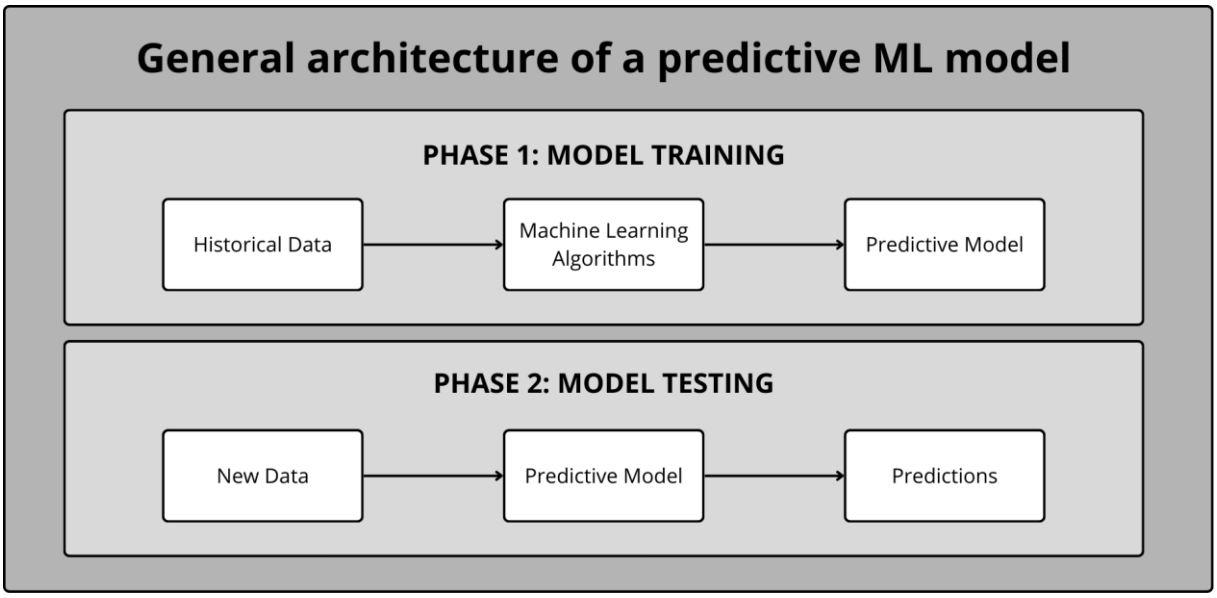


Figure 3: The general architecture of a predictive ML-based model, incorporating training and testing phases. Adapted from Sarker (2021).

Supervised learning aims to develop a function that maps inputs to outputs using labeled training data (Alam et al., 2025; Sarker, 2021). Thus, supervised learning can be used when an output is associated with specific inputs. As viewed in Figure 2, supervised machine learning is the preferred choice in this study. This is because the model is trained on historical data where both inputs and the target variable (workload) are known, allowing it to learn the relationship and make predictions. In contrast, unsupervised learning lacks a target variable and is used to display patterns in unlabeled data, semi-supervised learning combines limited labeled data with unlabeled data, and reinforcement learning learns through trial-and-error interactions with an environment (suitable for training AI models) rather than from historical labeled datasets (Alam et al., 2025; Sarker, 2021).

According to Sarker (2021), the most prevailing types of supervised learning tasks are "classification" and "regression". The main distinction between a "classification" and "regression" learning task is that "regression" attempts to predict specific numerical quantities or amounts (continuous target variable), while "classification" focuses on identifying which group an observation belongs to (discrete target variable) (Sarker, 2021). For instance, in a regression task, a model might predict the exact sales volume for a specific product by analyzing historical demand patterns and seasonal trends. Since the target variable (workload) is continuous in this study, supervised regression ML-based models are well-suited for this type of time-series forecasting problem. To evaluate the performance of forecasting models, error metrics must be used. When evaluating ML-based models, a benchmark against a simple baseline should be used to view the

results in a comprehensible way (Hewamalage et al., 2023; Petropoulos et al., 2022). In addition, error metrics should be selected based on data characteristics and forecasting circumstances, as no globally accepted error metric exists for all scenarios (Hewamalage et al., 2023).

Typically, forecasts in the context of production planning, workforce planning, and supply chain are based on statistical methods, management judgement, and machine learning modeling (Petropoulos et al., 2022). The forecasting approach using ML-based modeling positions the study within the domain of modern methods rather than traditional ones. Previous research has used ML-based methods to forecast workload within the manufacturing industry. In particular, De Simone et al. (2025) sought to identify the best model for predicting task workload using available historical data. However, such efforts have relied on limited datasets and have called for studies with larger, more diverse datasets (De Simone et al., 2025). Additionally, De Simone et al.'s (2025) paper focuses on technical production characteristics without considering social elements, which affect workload forecasting. In this study, the STS perspective is used to investigate both social and technical elements, as well as the technical implementation of administrative workload forecasting in complex, customization-intensive manufacturing environments. To apply ML-based models to available data, the data must be prepared.

2.4 Data mining

As organizations accumulate increasing volumes of data, data mining has emerged as a prominent central approach for converting raw data into actionable insights (Han et al., 2012). At its core, data mining is the process of extracting implicit, previously unknown, and potentially useful patterns from large amounts of data and converting them into knowledge (Han et al., 2012). For instance, data mining can be applied in organizational contexts to classify customer responses or predict continuous outcomes such as revenue, based on patterns identified in historical data (Han et al., 2012). In this process, ML algorithms provide tools that enable systems to learn from patterns in historical data and apply them to new observations (Han et al., 2012; Sarker, 2021). When applied to organizational contexts, the data mining process is typically structured through standardized frameworks such as CRISP-DM (Chapman et al., 2000; Schröer et al., 2021), with ML algorithms serving as the primary analytical tools for uncovering patterns in historical data (Han et al., 2012). To sustain a structured methodology, the CRISP-DM framework is used to process data and develop forecasts in this study.

2.5 Conceptual framework

To establish a foundational standpoint for this study, we developed a conceptual framework illustrated in Figure 4.

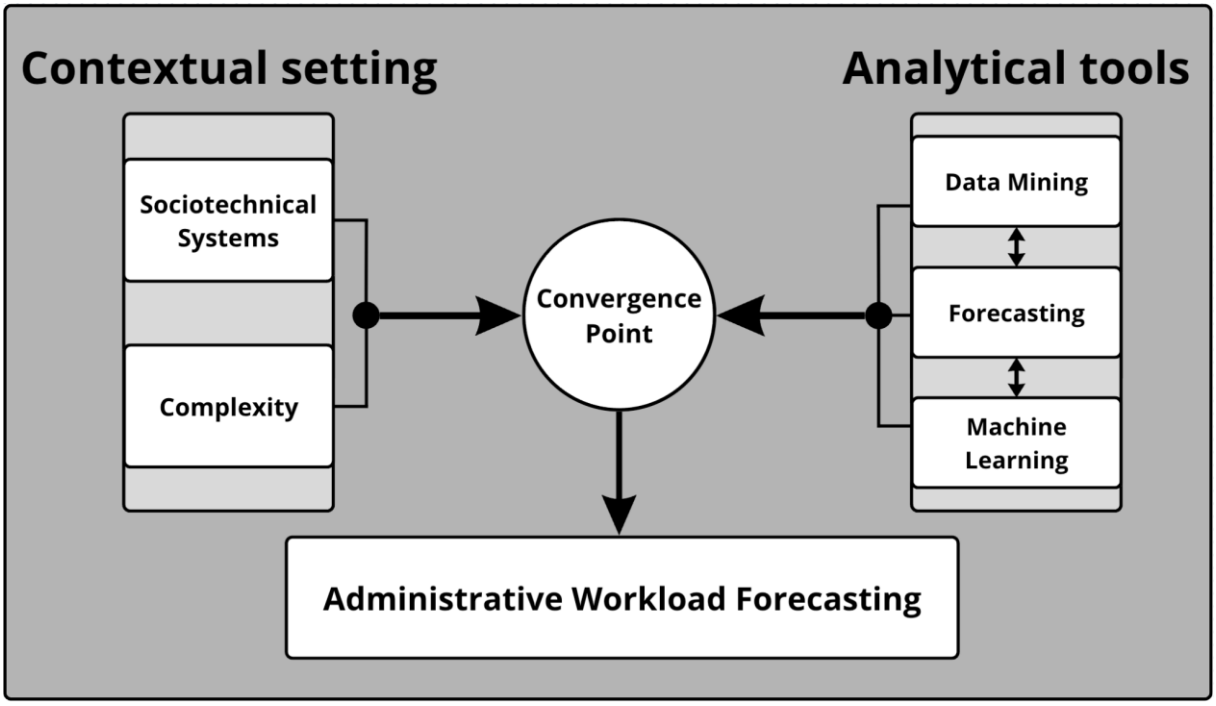


Figure 4: Conceptual framework.

Given the complex, customization-intensive manufacturing environment, the STS perspective was adopted, enabling examination of the CA department and its operations. By implementing the STS perspective, the CA department and its operations can be broken down into their core elements, providing insights into how administrative workload is created. Although the STS perspective serves as the organizational lens, it does not account for the non-linear interactions that arise when multiple elements interact simultaneously and repeatedly. Therefore, complexity contextualizes how these interactions produce uncertain outcomes within the environment in which the CA department operates. Together, the STS perspective and complexity provide the context for this study. The developed contextual setting enables the empirical data to be analyzed and to answer RQ1: how is administrative workload generated within a complex, customization-intensive manufacturing environment?

Furthermore, this unpredictability creates a need for organizations to anticipate future workload by drawing insights from available data. As a result, data mining is the process of extracting patterns from historical data. These hidden patterns serve as the foundation for forecasts. Especially valuable in this regard is ML, with its extensive ability to identify complex, non-linear patterns in historical data that would otherwise be difficult to detect. In particular, the branch suitable for this context is supervised learning, where a model is given labeled historical data to learn the relationship between input variables and a target output. The literature on analytical tools helps inform the choice of forecast approach and the analysis of quantitative empirical data. Thus, it enables answering RQ2: how does the predictive performance of different forecasting methods compare when applied to administrative workload data in a complex, customization-intensive manufacturing environment?

Collectively, the theoretical chapter's contextual setting and analytical tools converge on a single point of understanding, which together move towards administrative workload forecasting as the study's objective.

3 Method

This chapter details the research design adopted to address the research questions, outlining the transition from qualitative context mapping to the quantitative evaluation of forecasting models. Furthermore, the chapter covers the contextual setting, ethical considerations, and the validity of the research design.

3.1 Research design

This study investigates how resource planning can be improved in complex, customization-intensive manufacturing environments by integrating an understanding of administrative work practices with ML-based forecasting models. This is achieved by adopting an abductive single-case study setup. As an empirical method, case studies enable in-depth investigations of phenomena in their real-world contexts (Yin, 2018), making them suitable for mapping a complex organization. Case studies are flexible in the selection and combination of different sources of data, for instance, interviews, archival records, and observations (Eisenhardt, 1989; Yin, 2018), which are utilized in this study through the combination of both qualitative and quantitative data. An abductive research strategy enables alignment of emerging empirical insights with the theoretical framework iteratively throughout the research project (Bell et al., 2022), contributing to understanding the complexity within the CA department. Understanding of an empirical phenomenon and theory can be expanded by iterating back and forth among research techniques, empirical findings, and theory (Dubois & Gadde, 2002).

To address the case context and its complexity, an exploratory sequential mixed methods research design was developed. The study design draws on Schoonenboom and Johnson's (2017) guidelines for developing MMR studies. The purpose of the research design is to study the case context from a broader perspective, which an MMR design advantageously allows (Johnson & Onwuegbuzie, 2004). The initial qualitative phase maps and investigates how the CA department and its administrative activities can be viewed from an STS perspective. The insights from the qualitative phase inform the applicability of forecasting tools in the quantitative phase and facilitate the explanation of the quantitative results. The research design is appropriate for the purpose and case context since MMR respects and values insights from multiple research paradigms and contributes to the recognition and examination of complex settings (Johnson, 2017). Figure 5 visualizes the research design and its connection to the purpose and the research questions.

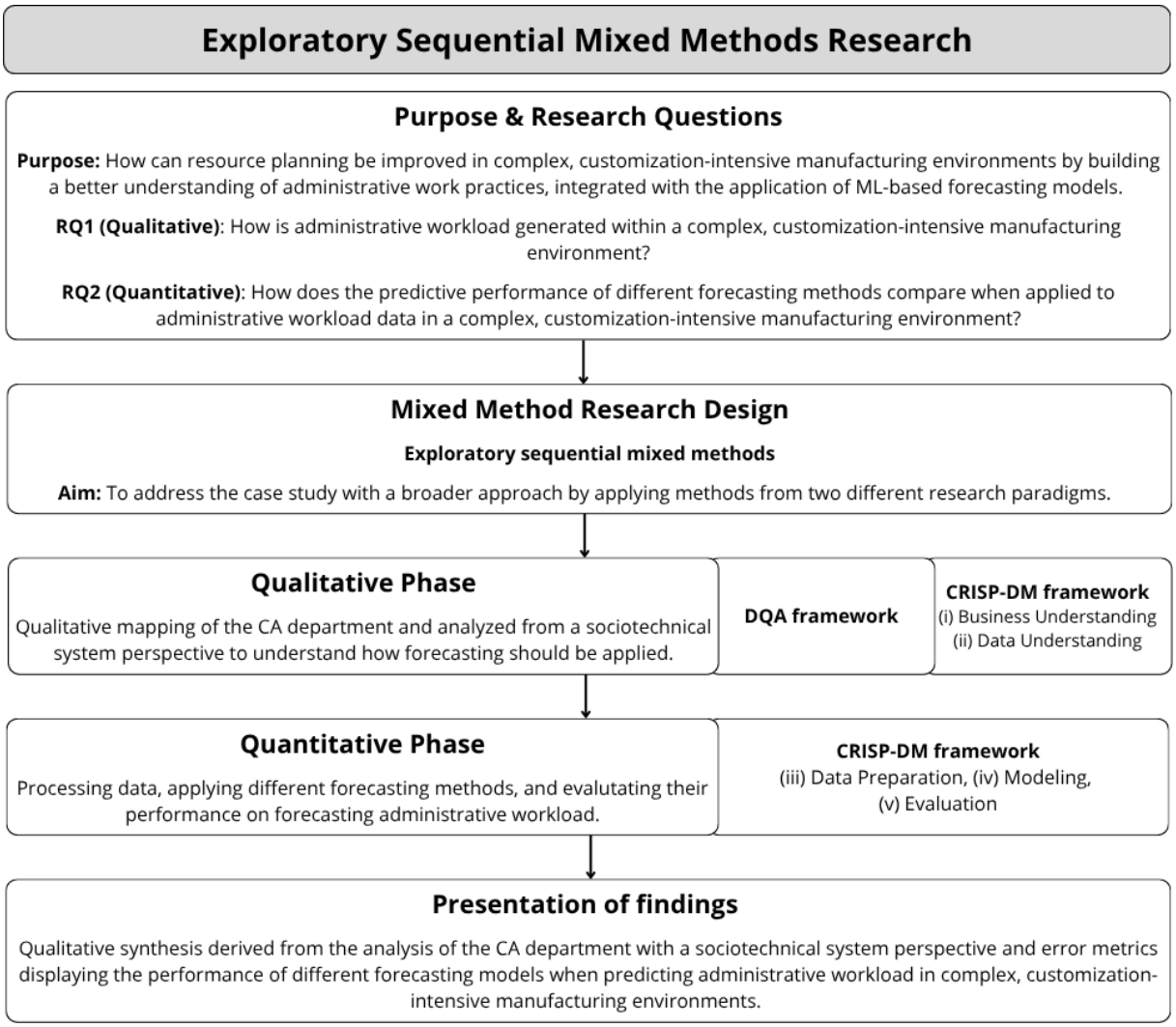


Figure 5: Exploratory sequential mixed methods research design.

3.1.1 The DQA framework

Data collected from the qualitative phase were systematically analyzed using deductive coding by following the steps of Fife and Gossner's (2024) framework for deductive qualitative analysis (DQA). The steps of DQA are explained in Table 1.

Table 1: Step-by-step guide for deductive qualitative analysis (Fife & Gossner, 2024).

Activity	**Description of activity**
1. Developing the research question and selecting a guiding theory	A clear research question is formulated by linking the phenomenon of interest to a relevant guiding theory, which provides the study's conceptual lens.

2. Operationalizing the theory	Key concepts from the guiding theory are translated into sensitizing constructs that can be used analytically throughout the study.
3. Gathering a purposive sample	Participants or data sources are strategically selected to ensure sufficient variation and depth to meaningfully apply the guiding theory.
4. Coding and analyzing the data	Data are analyzed through deductive coding based on the theory. Inductive coding can be applied to capture emerging insights to complement current theory.
5. Theorizing	Findings are synthesized by evaluating supporting, contradicting, refining, and expanding evidence, resulting in a revised or strengthened version of the guiding theory.

In accordance with the DQA framework, we have used RQ1 and the guiding theory (i.e., the STS perspective) as a basis for the qualitative phase. The STS perspective was identified as the most appropriate approach for studying the empirical context of the case company, as it allows both social and technical aspects to be taken into account (Bostrom & Heinen, 1977; Kemp et al., 2024). By applying DQA, the existing theory can be operationalized as sensitizing constructs (Fife & Gossner, 2024). The sensitizing constructs consisted of the four elements of the STS perspective: (i) Structure and (ii) People, representing the social system, and (iii) Physical System and (iv) Task, representing the technical system (see Figure 1).

3.1.2 The CRISP-DM framework

The study used the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework as the methodological basis for developing the forecasting tool. The process framework is widely used and standardized for initiating data analytics projects, both in research and across industries (Schröer et al., 2021). The framework is commonly employed in studies concerning forecast implementations, for instance, De Simone et al. (2025). The process framework consists of the following six phases: (i) Business Understanding, (ii) Data Understanding, (iii) Data Preparation, (iv) Modeling, (v) Evaluation, and (vi) Deployment (Chapman et al., 2000).

While the six original phases were followed, minor adaptations of the framework were made to align it to the case context. Notably, the Business and Data Understanding phases were conducted prior to the quantitative phase (Data Preparation, Modeling, and Evaluation) through the qualitative case study work. Furthermore, the Deployment phase was limited to model recommendation in the thesis. An overview of the method is shown in Figure 6.

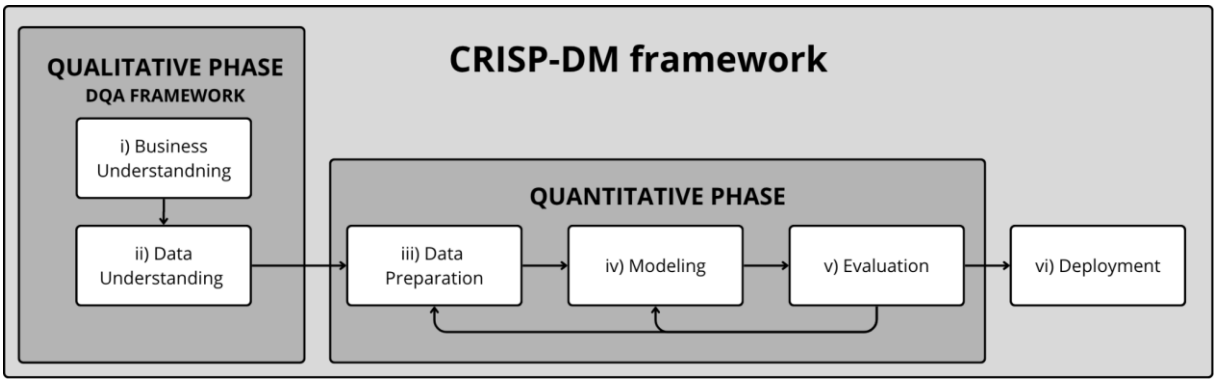


Figure 6: CRISP-DM framework adapted for the case study.

3.2 Contextual setting

The case study was conducted at a global truck manufacturer, focusing on a facility in Sweden. The facility serves an international business-to-business market with trucks through its product range, spanning long-haul transportation, regional distribution, construction operations, and specialized applications.

The plant follows a make-to-order logic where customers, together with sales representatives, create the specification for the desired truck. All orders containing standardized components and options are classified as standard orders and follow the standard production flow. In addition, the company offers a high degree of customization based on customer requirements. These orders are classified as CA orders and require additional administrative processing before the orders are prepared for production. This production logic corresponds to a design-to-order approach. From March 2024 to March 2026, the average proportion of orders with customization was 57%, with monthly values ranging from 46.7% to 70.8%.

To manage the high degree of customization, the CA department handles administrative tasks related to CA requests and ensures that modified trucks can be produced. The CA department constitutes a complex network of flows and administrative tasks, making it difficult to oversee. Its general structure can be divided into three main units: Offering, staffed by case managers, Design Engineering, staffed by design engineers, and Preparation Engineering, staffed by preparation engineers. Together, the units ensure that customization orders are manufacturable.

The CA department has no insight into future administrative workloads for processing CA orders. The lack of insight complicates human capacity planning in the CA department and the ability to plan CA orders in the production flow as efficiently as standard orders, which affects the delivery date provided to the customer. However, historical data exists for administrative tasks within the CA department. Yet, the data has not been used to generate insights due to the department's complexity. The CA department, together with its data, presented an opportunity to understand complex, customization-intensive manufacturing environments and forecast future administrative workload. Understanding administrative tasks in the CA department and insights into future workload will improve managers' planning ability and contribute to better working conditions for employees, as fewer unforeseen surges in workload will affect the CA department.

For these reasons, the CA department is a suitable venue for mapping a complex system and applying and evaluating the applicability of various forecasting methods for administrative workload in complex, customization-intensive manufacturing environments.

3.3 Ethical considerations

Conducting responsible research requires careful attention to ethical considerations. To ensure ethical compliance, the four principles of Bell et al. (2022) were followed, namely: avoidance of harm, obtaining informed consent, protection of privacy through confidentiality, and preventing deception.

As the study concerns administrative workload, it was explicitly made clear to all participants that the purpose was not to evaluate individual performance. This framing was intended to prevent stress and mitigate any risk of professional or reputational harm. Moreover, physical safety was ensured by complying with the manufacturing facility's safety regulations throughout the study period. Regarding informed consent, participants were briefed on the study's purpose prior to each interview and observation session, and were informed that participation was voluntary. No individual was pressured to participate, and the option to withdraw was communicated clearly. To protect privacy, participants, the case company, and associated data were anonymized where appropriate. The collected qualitative data were deleted after the completion of the thesis. Data was handled exclusively on the case company's devices, where they were stored securely and processed in accordance with GDPR and the company's data policies. Finally, to prevent deception, the scope and purpose of this study were communicated transparently to all participants, and the intended use of the collected data was outlined before the interviews or observation sessions started.

3.4 Validity of the research design

According to Onwuegbuzie and Johnson (2006), assessing the validity of an MMR is complex because it requires addressing the quality of both the quantitative and qualitative components, as well as their integration. To address this complexity, a legitimation framework was developed to justify the research design and account for inherent tensions arising from combining the two paradigms (Onwuegbuzie & Johnson, 2006; Schoonenboom & Johnson, 2017). Table 2 outlines the nine dimensions of validity addressed in this thesis, which are derived from the methodological frameworks of Onwuegbuzie and Johnson (2006), Johnson and Schoonenboom (2016), and Schoonenboom and Johnson (2017). By systematically addressing multiple validity dimensions within an MMR design, the study ensures an integrated, well-founded analysis. This collectively strengthens the study's credibility and thereby achieves legitimacy.

Table 2: Legitimation of the mixed methods research design (Onwuegbuzie & Johnson, 2006; Johnson & Schoonenboom, 2016; Schoonenboom & Johnson, 2017).

Validity dimension	**Definition**	**Application in the study**
Inside-Outside	The degree to which the study accurately captures and balances participants' subjective perspectives with the researcher's objective observations.	Inside perspectives were captured through unstructured interviews and participant observations across all units and roles of the CA department. Triangulation was achieved through interviews, observations, and document analysis, which strengthened the balance between participant perspectives and our own observations.
Weakness Minimization	How well the combined use of qualitative and quantitative methods compensates for the individual weaknesses inherent in each approach.	The quantitative phase tests model performance across multiple markets. To compensate for the inability of quantitative methods to explain the conditions under which workload is generated, the qualitative phase mapped the CA department through interviews, observations, and document analysis.
Sequences of Activities	The extent to which findings from preceding research phases are appropriately incorporated into and inform subsequent phases.	The qualitative phase identified the target variable, aggregation level, input variables, and relevant systems with data, used in the subsequent quantitative phase (see section 4.4).
Conversion	The degree to which transformations applied to the data are sound, and the conclusions drawn from them are valid and well-grounded.	Qualitative data were coded deductively through the STS perspective using DQA. Quantitative data were aggregated to market-period level and validated before model training. The abductive approach enabled iterations between empirical findings and literature throughout both phases.
Sample Integration	The degree to which findings derived from the combined samples support the	The two phases in the study do not address the same sample. Instead, they shed light on the studied case environment from different

	conclusions made across both research phases.	perspectives by presenting quantitative forecast results and the administrative workload generated, which contribute to the conclusion.
Data integration	The degree to which a unified and consistent understanding emerges from the combination of data, analysis, and conclusions across both phases.	Qualitative findings informed the data preparation for the quantitative phase. The qualitative findings are also used to explain patterns in the quantitative results.
Pragmatic	How well the mixed methods approach served to fulfill the overall purpose of the study.	The purpose of the study was achieved with the MMR approach, as the qualitative or quantitative methods alone are insufficient to address the forecasting of administrative hours in a complex, customization-intensive manufacturing environment.
Socio-political	The extent to which the research process acknowledges and accounts for the perspectives and interests of the different parties involved in or affected by the study.	Participants in different roles and at various decision-making levels were interviewed. The insights shaped the forecasting tool's business objective: to predict future administrative workloads. This was in the interests of both managers and employees, as it can improve the work environment and planning ability. In addition, we ensured that the forecasting tool could not measure individual performance.
Multiple Validities	The degree to which all relevant validity dimensions have been explicitly identified and adequately addressed throughout the study.	Multiple validities are ensured through validity assurance in the separate studies before overall conclusions are drawn. By following methodological frameworks in both phases, data collection and analysis can be carried out with high quality, laying the foundation for validated overall conclusions.

4 The qualitative phase

This chapter outlines how the study was executed in the qualitative phase and presents the findings obtained. In order to answer RQ1, the CA department was mapped out using interviews, participant observations, and documentation. Furthermore, interactions between key elements and their generation of administrative workload were analyzed.

4.1 Data collection

During the qualitative phase, data were collected to develop an understanding of the CA department, its technical systems, its administrative processes, and the quantitative data available for forecasting administrative workload. In total, eight unstructured interviews were conducted (13.25 hours), four participant observations (six hours), and ten internal documents were studied (see Table 3). Interviews were not audio-recorded; instead, the collected data consisted of notes and drawings. The data collection phase for the unstructured interviews took place between weeks four and seven, while the review of documents and observations continued throughout the study. Interviews were the primary method, as they are useful for discovering how and why activities are performed in a certain way (Yin, 2018).

Given the minimal prior knowledge of the CA department and the relevant themes in the literature for their setting, unstructured interviews were deemed most appropriate (Qu & Dumay, 2011). This was partly because a prepared questionnaire risked focusing on unimportant areas, and partly because the interview participants worked with widely varying tasks, meaning that a questionnaire would not be relevant to everyone.

Table 3: Data collection in the qualitative phase.

Data Source	**Unit**	**Collected Data**	**Collection Method**	**Data volume**	**Purpose**
Interviews	Production planning department, Sales department, CA department	Notes and drawings	Unstructured interviews	13.25 hours	Understand the administrative work in the CA department and its available data
Observations	CA department, CA shop-floor	Notes	Participant observations and weekly on-site observations	Six hours + weekly on-site observations	Understand the usage of different systems, department culture, and department interconnections

Documentation	CA department, Production planning department	Introductory manuals, user guides, statistics	Document review	Ten documents	Reinforce the understanding of the case study

4.1.1 Choice of participants

Participants for the unstructured interviews were selected through purposive sampling across the different roles in the sales organization and the CA department. Purposive sampling is used to capture process variance and multiple perspectives (Bell et al., 2022). Experienced employees (i.e., employees who have worked more than three years in the CA department) were engaged in the unstructured interviews to gain insights about the process flow, daily activities, frequently used systems, and communication touchpoints.

Representatives from four departments (Sales, CA, CA shop-floor, and Production planning) participated in the unstructured interviews. Within the CA department, the following employees were involved: four case managers from the Offering unit, one design engineer from the Design Engineering unit, and one preparation engineer from the Preparation Engineering unit. To understand the administrative activities within the CA department, knowledge of the incoming and outgoing workload, as well as internal activities, was important.

4.1.2 Unstructured interviews

The complexity of the CA department, with its ambiguous interfaces, processes, and relationships, made it difficult to predict relevant themes, which rendered a predefined interview guide inappropriate. Unstructured interviews were therefore the preferred option for studying a complex setting, as they allow the interviewee to articulate and structure their own perspectives on the phenomenon under investigation (Mulcahy et al., 2021). The interviews began with the question "How do you perform your work?", after which the participant explained and elaborated, and we followed up on topics that seemed relevant, which is common practice in unstructured interviews (Bell et al., 2022). This allowed the in-depth experiences of employees to be captured and the CA department to be mapped in an exploratory manner. Thus, the method for the data collection was adapted to the setting and the pursued outcome, significant for abductive case studies (Dubois & Gadde, 2002).

All interviews were conducted in person on-site at the case company, and a total of eight interviews were completed. Details regarding the interviews are presented in Table 4.

Table 4: Interview details.

Interviewee index	Participant / Unit	Data type	Date	Length	Purpose
MPP1	Manager Production Planning	Text notes	2026-01-20	3 hours	Understand the planning process of CA orders
PP1	Production Planner	Text notes	2026-02-22	1 hour	Understand the planning process of CA orders
SD1	Sales Department	Text notes and drawings	2026-01-23	1.5 hours	Understand the inflow of work to the CA department
CM1	Case Manager Offering	Text notes and drawings	2026-01-26	3 hours	Understand the administrative work in the Offering unit and its available data
DE1	Design Engineer	Text notes and drawings	2026-01-27	2 hours	Understand the administrative work in the Design Engineering unit and its available data
PE1	Preparation Engineer	Text notes and drawings	2026-01-28	1 hour	Understand the administrative work in the Preparation Engineering unit and its available data
CM2 & CM3	Case Managers Offering	Text notes and drawings	2026-01-29	1 hour	Understand the administrative work in the Offering unit and its available data
CM4	Case Manager Offering and Key user to Order Manager	Text notes	2026-02-03	0.75 hours	Understand the Order Management system and what data it could provide for a forecasting tool

4.1.3 Documentation

Internal documentation was examined to supplement interviews and observations, thereby reinforcing understanding of the case study. The documentation studied includes introductory manuals for positions in the CA department, user guides for existing forecasting tools, business intelligence dashboards, historical statistics for CA orders, and administrative occupancy statistics. In total, ten documents were studied.

4.1.4 Participant observations

Participant observations were made during the study, where work processes and CA employees' interactions with different systems were observed. This is to fill knowledge gaps that have not been covered by other data collection methods (Yin, 2018). Occasionally, we observed and replicated tasks in systems to better understand the administrative work within the CA department. This was carried out by actively participating in administrative tasks, such as processing incoming orders in the order management system, while the participants explained the work and answered questions as they arose. Insights from participant observations were documented through text notes and drawings depicting the interconnections between the units and systems in the CA department. Participant observation enabled the study to see administrative tasks from a user's perspective, a benefit noted by Yin (2018). Furthermore, during meetings, participant observation sheds light on the department's culture and the interconnections between the CA department and other departments. Participant observation was conducted for a total of 6 hours during the study. In addition, direct observations were conducted weekly while working on-site at the case company.

4.2 Data analysis

The qualitative data were coded and analyzed through a sociotechnical lens, with empirical evidence identified in relation to the theory in accordance with the DQA framework (Fife & Gossner, 2024). When the collected data were coded, the four elements of the STS perspective served as guiding themes for categorizing the codes. Text notes and drawings from the interviews and observations were coded individually and subsequently merged together by theme. Finally, interconnections between the four elements were analyzed. The analysis method enabled us to structure the collected unstructured data.

As a final step of DQA, syntheses were created from coded data, providing insights into how the CA department can be mapped and how administrative work in the CA department is generated through interactions among the elements in STS. However, the purpose of using DQA in the analysis is not mere theorization but the reconceptualization of sociotechnical concepts when studying administrative tasks in a complex, customization-intensive manufacturing environment. By applying the STS perspective to the case, a deeper understanding of administrative workload in the complex, customization-intensive manufacturing environment can be gained.

4.3 Findings

The following section presents the findings from the qualitative phase, focused on mapping out the CA department and the STS-based analysis of administrative workload. Furthermore, the Business and Data Understanding is presented, which is the foundation for the development of the forecasting models.

4.3.1 The CA department mapped out

Mapping of the CA department revealed a sequential workflow in which incoming CA requests from the customer are processed before reaching the CA shop-floor for assembly. The workflow, but also the different units within the CA department, are visualized in Figure 7.

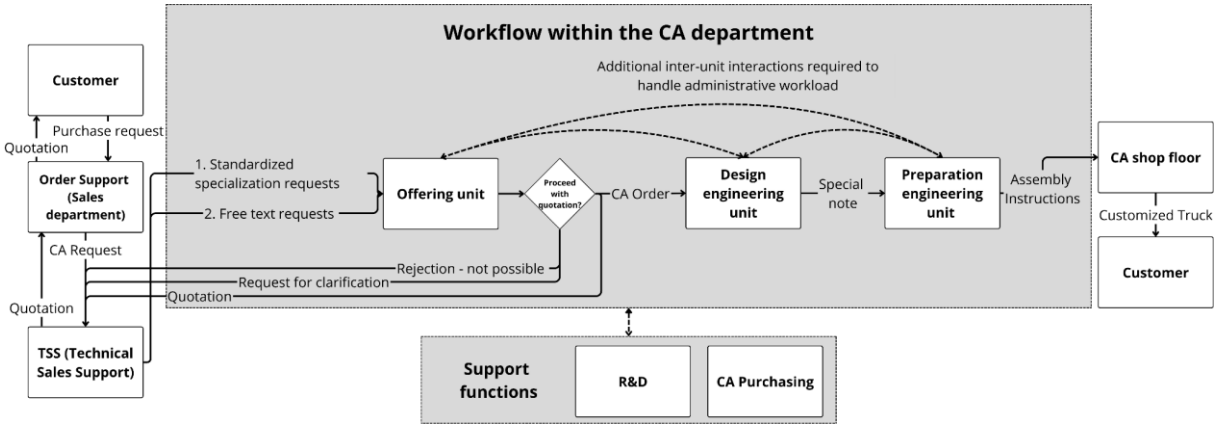


Figure 7: The CA department mapped out.

The initial entry point within the CA department is the Offering unit, staffed by case managers. These case managers assess the feasibility of incoming CA requests, provide quotations to customers, and transfer CA orders to the Design Engineering unit. To clarify, feasibility evaluation requires determining whether a request is both legally permissible and physically possible, which is mandatory for all incoming CA requests.

Some incoming requests follow a structured specification format, providing case managers with sufficient detail for direct evaluation. Others are submitted as free-text requests, requiring interpretation of the intended solution before assessment. Upon receipt, case managers evaluate each CA request and determine one of three outcomes. Firstly, if the request contains adequate detail and the proposed solution is feasible, it is converted into a CA order and transferred to the Design Engineering unit. Secondly, if the request lacks sufficient information or is ambiguous, clarification is requested from the sales department before evaluation. Thirdly, if the request is deemed infeasible, it is rejected and returned to the sales department. However, if a request exceeds current knowledge and presents an interesting business opportunity, the Research and Development (R&D) department may be consulted before a response is sent to the Sales department.

Upon receiving a CA order, the design engineer reviews the requested modifications and produces a design specification detailing the final design. The design specification is subsequently forwarded to the Preparation Engineering unit. Simultaneously, the CA department is supported by CA

Purchasing, which procures necessary parts and monitors them throughout the process. Meanwhile, a preparation engineer reviews the design specification and creates assembly instructions needed for production. Lastly, these completed instructions are forwarded to the CA shop-floor, where the CA order is manufactured. Moreover, the dashed lines between units show additional inter-unit interactions necessary to handle the administrative workload. This involves communicating capacity issues, errors, and a lack of information.

4.3.2 Findings from DQA

The CA department is regarded by managers as a strategically vital unit that enables the organization to differentiate itself from competitors. In this light, the CA department processes a significant share of all truck orders. However, the CA department operates without an established limit on incoming orders, resulting in incoming work volumes exceeding its processing capacity. In addition, the delivery timeframes are set by the sales department before the CA department receives the order. This leaves the CA department with limited control over when and how much work arrives.

Employees experience persistent time pressure due to the high volume of incoming work and the sales department's delivery timeframes. As a consequence, employees reported having little to no time for self-reflection or professional development. Participant CM1, for instance, stated that there is rarely any time outside daily work tasks to focus on skills that would further their professional growth. Moreover, participant DE1 stated that a substantial portion of working time is spent correcting errors from previous administrative steps. These corrections displace the work employees are expected to complete within their set deadlines. It was also reported that other colleagues' work was regularly checked out of concern that errors would remain undetected. This behavior was noted across multiple roles and units across the CA department. From a managerial level, it was reported that the CA department lacks visibility into its upcoming workload. Thus, managers have no structured means to anticipate either volume or the complexity of future requests. As participant MPP1 expressed[1]: "We are blind to future workload for the moment."

A recurrent theme regarding tasks was that high order variability often meant tasks were rarely repeated in the same form. Participant DE1 asserted that various technical dependencies affected the CA orders, making them unpredictable, as the same solution concept was not always applicable across different truck models. Furthermore, Participant DE1 noted that the design systems do not automatically enforce technical restrictions, so design engineers must rely on their own expertise or consult reference documents to ensure that modifications are feasible. The CA request formats were reported as a contributor to this variability, as some inquiries were submitted in a structured format, while others arrived as free-text requests that required interpretation and often several clarifications from the sales department before the correct solution could be established.

Furthermore, rework was identified as a regular task of the daily workload. In particular, participant PE1 noted that late changes to a CA order required units to redo their work entirely, with entire

[1] Since no interviews were recorded, only one quote was used, as it was noted word-for-word. The remaining references to interviews were instead presented through paraphrases.

working weeks sometimes consumed by rework. Although the standard practice is that preparation engineers should receive the design at least two weeks before the order reaches its definitive state, after which no further modifications can be made, this was regularly not met. As a consequence, the CA department's units have to reprioritize their workloads at short notice, which, in some cases, leads to delays in truck production.

Several systems support the administrative work in the CA department, and different units and roles rely on different systems to perform their daily tasks. However, it was noted that data across these systems is uneven. Participant CM4 clarified that not all data is available across all systems, nor do all roles and units have the same access to the same data. This means that information availability depends on roles and positions within the workflow. Alongside established systems, temporary solutions were also observed. These temporary solutions complemented activities that the established systems could not cover. Participant CM1 brought forward one such example: a temporary list for storing CA orders that exceed the Design Engineering unit's regular capacity until they can be processed further. Furthermore, while a forecasting tool for operational hours exists, no equivalent system for administrative hours within the CA department exists. As such, this absence means the technical infrastructure supports shop-floor workload planning for production but not the administrative work required to prepare orders for production.

4.3.3 Business understanding

Through interviews, observations, and document studies, a comprehensive understanding was obtained of the organizational structure of the CA department (see Figure 7 in section 4.3.1), the administrative processes related to CA orders, and the identification of challenges related to administrative workload forecasting. The qualitative findings revealed that no current tool exists to support managers with resource planning within the CA department. This has led to limited insight into the CA department's upcoming workload. The business problem was therefore defined as: the development of a forecasting tool that provides useful insights into the administrative workload ahead for managers in the CA department.

When the business objective for the CA department was understood, and the target variable was identified (administrative workload in hours), the data mining task could be specified as a supervised regression task, as the target variable is continuous. The data mining objective could consequently be defined as follows: To develop and evaluate forecasting models capable of predicting administrative workload per market and period using historical order data.

4.3.4 Data understanding

The interviews and observations were likewise employed to gain insight into which systems stored relevant data, how administrative workload was recorded, how order characteristics were structured, and how reliable the data was. The data understanding ensured that the selection of data attributes was grounded in the case context, the forecast model reflected actual organizational needs (obtaining insights into future administrative workload), and the interpretation of data was aligned with how the operations in the CA department are carried out. The data consisted of archive data extracted from order management systems and a production planning database that

stored information about previous orders. The data sources and the data obtained from them are described in Table 5.

Table 5: Overview of the data sources.

Data source	Data amount	Time span	Description
CAQ (Customer Adaptation Quotation)	95 256 rows	January 2022 to February 2026	Historical data divided by the units within the CA department containing: ● Administrative hours used for a customization request. ● Assembly date for the trucks. ● Which market the truck is sold to.
OPT - (Order Planning Tool)	90 841 rows	April 2024 to February 2026	Historical data containing: ● Total number of orders ● Number of CA orders ● Assembly dates for the trucks. ● Which market the truck is sold to.

However, the data sources contained more data attributes than those listed in Table 5, but these were removed as they were not relevant for forecasting future workloads in the CA department. A data attribute was retained as an input variable only if it was relevant for forecasting future workload and would be available at the time of forecasting. Since the forecast can only be used if the input data resembles and is formatted exactly like the data it was trained on, the selection of input variables was limited. The modeling variables retained for their relevance to forecasting administrative workload (the target variable) included year, period (the case organization's method of representing months), market, and number of trucks with CA jobs.

4.4 Qualitative findings informing the quantitative phase

The qualitative phase informed the quantitative phase by providing an understanding of the CA department's operations and data availability. More specifically, the mapping of the CA department's administrative work practices and the DQA findings informed the subsequent CRISP-DM phases (Data Preparation, Modeling, and Evaluation) in five ways. Firstly, the target variable was set as administrative hours per market and period. Secondly, the qualitative findings determined the level of data aggregation to market-period, reflecting how the CA department organizes and monitors its work. Thirdly, the choice of input variables was limited to data attributes that would be available at the time of the forecast, which was identified through interviews and observations. Fourthly, systems that stored relevant data were identified, resulting in the extraction of historical data from the CAQ and OPT data sources for use in the quantitative phase. Finally,

the qualitative understanding of how administrative workload is generated within the CA department formed the basis for interpreting the quantitative results in the discussion.

5 The quantitative phase

This chapter outlines the execution of the study in the quantitative phase. Additionally, it presents the hypothesis and the results obtained. In order to answer RQ2, we prepared the data from the case company for model training, trained ML-based models, and evaluated their performance.

5.1 Hypothesis development

Findings from the qualitative phase indicate that administrative workload emerges through interactions between system elements in a non-linear manner, suggesting that such patterns are difficult to capture using simple linear relationships. Previous research argues that ML-based models are easier to adapt to data of different characteristics to capture their patterns (Petropoulos et al., 2022). Including intuitive baseline models, such as Simple Average, to benchmark forecasting performance is also highlighted by previous research as important (Hewamalage et al., 2023; Petropoulos et al., 2022). Thus, the following hypothesis is formulated:

Hypothesis: Non-linear (ensemble-based) ML models show higher predictive accuracy than simple baseline models, as well as linear and polynomial regression models, when forecasting administrative workload in complex, customization-intensive manufacturing environments.

5.2 Data preparation

In the Data Preparation phase, the initial raw data were transformed into the final dataset required for modeling. The phase included data integration, cleaning, transformation of datasets, and attribute construction, in accordance with the activities outlined in the CRISP-DM framework (Chapman et al., 2000).

In the case context, data preparation was necessary due to the use of multiple raw data sources and the need to structure the dataset in an aggregated format (records aggregated by market and period) to address the forecasting goal (administrative workload per market and period). The aggregated structure preserves both variation over time and across markets, allowing the model to capture differences between markets.

The data used in the study originated from two internal systems (see Table 5 in section 4.3.4). Since the data was retrieved in separate Excel files, integration was required. The first dataset with administrative hours and the second dataset with the number of orders with CA jobs were first aggregated by market and period. The datasets were then merged, using the market and period as the joint denominator. This enabled forecasting on the desired level, where each observation represented one market in one period. Thus, the unit of analysis became market-period.

During the integration phase, data cleaning was performed to address inconsistencies and quality issues. To adress the inconsistencies and quality issues, duplicate observations were removed, missing values were replaced with the correct value or the whole row was removed, variables not available at the time of forecast application were excluded, and aggregated results per market and period were verified. The choice of explanatory variables was constrained by the availability of those variables at the time of forecasting. This constraint ensured that a trained model could be

realistically implemented in reality without relying on unavailable information in real-time planning situations.

Based on the qualitative insights and data availability constraints, the following attributes were retained for modeling:

- Year
- Period
- Market
- Number of trucks with CA jobs
- Administrative hours per market and period (target variable)

These historical values reflect past CA volumes per market and period, forming the empirical basis for forecasting future administrative workload. This dataset formed the input for the subsequent Modeling phase. The data-refinement process is visualized in Figure 8. KNIME (version 5.3.3) was used as the data analysis tool to prepare data, train forecasting models, and test their performance. KNIME is used in the study due to its suitability for business intelligence and analytics, ease of use, and its interface that enables visualization and analysis (Hemlata & Gulia, 2019).

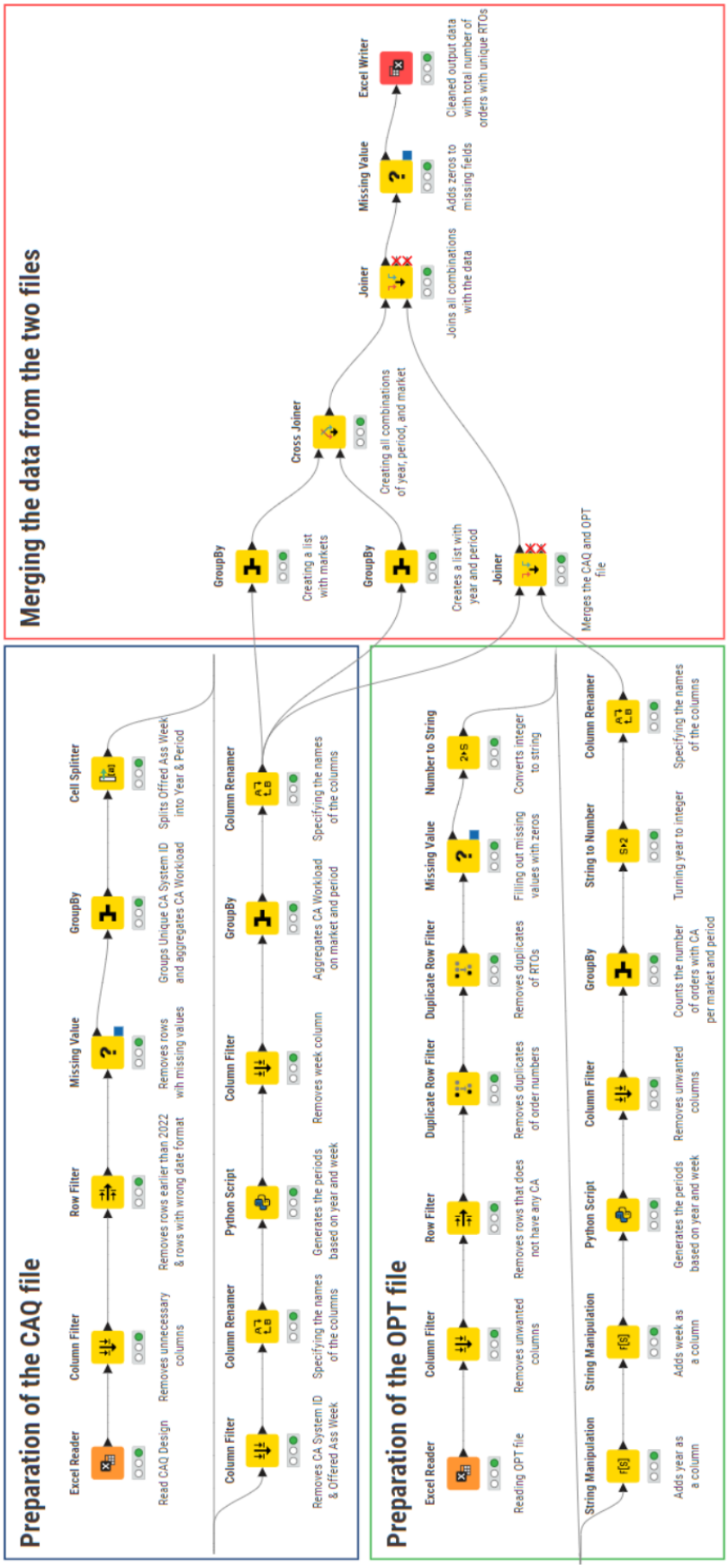


Figure 8: Preparation flow of data for the CAQ and OPT file in KNIME.

5.3 Modeling

In the Modeling phase, different forecasting models with the aim of predicting administrative workload per market and period were selected and trained, using the prepared dataset in accordance with the CRISP-DM framework (Chapman et al., 2000). Multiple regression techniques were selected as candidates for model construction. By distributing the choices across different regression techniques, it was possible to investigate which technique was best suited to the case context and its data. No final model selection was carried out in this phase. Instead, the model selection was done in the subsequent Evaluation phase.

In line with the CRISP-DM framework, the data was split into a training set and a test set (Chapman et al., 2000), using a time-based partitioning strategy. This was achieved by using all data from April 2024 to September 2025 (18 periods) to train the models, while data from October 2025 to February 2026 (five periods) was saved for model testing. This approach ensures that the model is trained exclusively on past observations and evaluated on future observations, reflecting the actual forecasting context in the case company. For the same reason, random partitioning was avoided to prevent the risk of data leakage, whereby future information could influence the training process.

In Table 6, the selected forecasting methods used in KNIME are listed and described. Different methods were selected to take into account both linear and non-linear relationships. The methods were selected due to their common use in forecasting and predictive modeling (e.g., De Simone et al., 2025) and their availability in KNIME.

Table 6: Description of forecasting methods (Sarker, 2021).

Forecasting method	**Description**
Multiple Linear Regression	A supervised machine learning technique that identifies a relationship between a continuous dependent variable and one or more independent variables. It uses a best-fit straight line to predict numerical target outcomes.
Polynomial Regression	Polynomial Regression models an nth degree polynomial relationship between independent and dependent variables.
Random Forest	Random Forest is an ensemble technique that fits multiple decision trees in parallel to improve accuracy. Generally, ensemble methods combine multiple decision trees to reduce errors and improve prediction stability. It uses a "bagging" strategy, training each tree on a different random subset of data. By averaging these individual results, the model minimizes over-fitting and provides more stable numerical predictions.

Gradient Boosted Tree	This ensemble learning algorithm generates a final predictive model from a sequence of individual models, typically decision trees. It utilizes the gradient to minimize a loss function. This process operates similarly to how neural networks use gradient descent to optimize internal weights for better performance.
XGBoost Tree Ensemble	XGBoost Tree Ensemble is an advanced, high-performance version of gradient boosting designed for speed and accuracy. It builds a sequence of trees where each new model focuses on correcting the errors of the previous ones. It includes built-in safeguards to prevent over-fitting, ensuring the model generalizes well to new data.
Simple Average	A baseline model (mean per market) was included to provide a reference point for predictive performance comparison with the other regression models.

Model training was conducted in KNIME using dedicated training nodes that correspond to each selected model technique. To maintain consistency, each model was trained on the same dataset with the same explanatory variables. The model parameters were configured using the recommended defaults. The training workflow is shown in Figure 9.

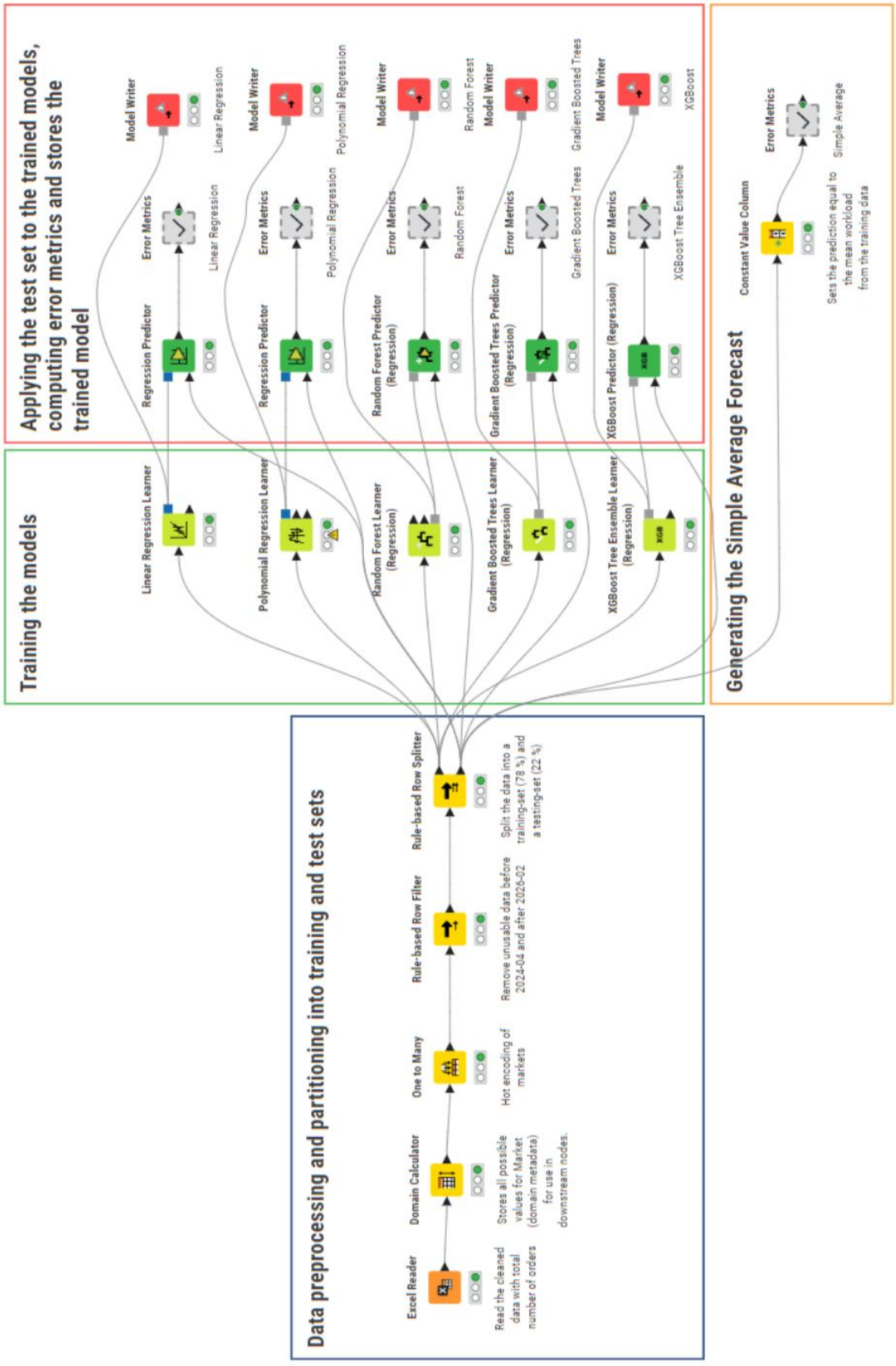


Figure 9: ML modeling flow in KNIME.

5.4 Evaluation

In this study, the Evaluation phase focused on assessing the predictive performance of the trained forecast models in relation to the data mining objective (Chapman et al., 2000). Determination of which model produced the most accurate estimates of administrative workload per market and period was in focus. To evaluate the predictive performance of the models, error metrics were applied to the data sets. Since opinions regarding the optimal error metric vary in the literature (e.g., Chicco et al., 2021; Hewamalage et al., 2023; Hyndman & Koehler, 2006), several were selected for use in the study to provide a broader perspective and capture different aspects of errors. The selected error metrics and their accompanying description are displayed in Table 7, and were selected on the same principle as the forecasting methods, namely, their frequent use in forecasting and availability in KNIME.

Table 7: Description of Error Metrics for measuring model performance (Chicco et al., 2021; Hewamalage et al., 2023; Hyndman & Koehler, 2006).

Error Metric	**Description**
MAE (Mean Absolute Error)	Measures the average of the absolute errors. It is a scale-dependent measure (i.e., the value depends on the scale of the data). MAE is resilient against outliers and may be preferred when communicating due to its simplicity.
sMAPE (Symmetric Mean Absolute Percentage Error)	Measures the average of the symmetric absolute percentage errors. Scale-independent measurement that allows for comparison of forecast performance across different datasets.
WAPE (Weighted Absolute Percentage Error)	Measures the total absolute forecast error divided by the total actual values, giving more weight to larger observations. Unlike sMAPE, WAPE aggregates errors across all observations before scaling. That makes it less sensitive to small actual values and more representative of datasets with high variation.
R^2	R^2 is the coefficient of determination and indicates the proportion of the variation in the dependent variable that can be explained by the model's independent variables. It is thus a measure of how well the model explains the relationship relative to the total variation in data.

The trained models were applied to the test datasets to generate estimates of administrative workload per period and market. The estimated values were subsequently compared with the actual values in the data set. Based on the calculated error metrics, the models were evaluated according

to their predictive performance. In addition to comparing the forecast models against each other, a baseline consisting of the historical mean administrative workload per market and period was also created and compared with the models' forecasted workload. The comparison made it possible to assess whether more advanced models provided a meaningful contribution beyond simple estimation methods.

5.5 Results

The quantitative results present the models' performance and assess the proposed hypothesis. Furthermore, a residual analysis for one model shows deviations in the predictions.

5.5.1 Model performance

Table 8 presents the results and error metrics from the workload prediction. The dataset was finalized as 1978 rows, after aggregation of data to achieve the desired unit of analysis (market-period). Thus, the models were trained on 1548 rows of data and tested on 430 rows. The best-performing models, which minimize MAE and WAPE and maximize R^2, are presented with italics and bold text.

Table 8: Results and error metrics for workload prediction

Dataset	**Method**	**MAE [h]**	**WAPE [%]**	**sMAPE [%]**	**R^2**
Train: 1548 Test: 430	Multiple Linear Regression	17.8	58.1	148.1	0.90
	Polynomial Regression	N/A	N/A	163.8	N/A
	Random Forest	***9.9***	***32.2***	***135.9***	***0.91***
	Gradient Boosted Trees	***9.9***	***32.4***	***135.7***	***0.93***
	XGBoost Tree Ensemble	11.2	36.6	137.7	0.90
	Simple Average	42.5	138.8	154.6	N/A

The results show that Random Forest and Gradient Boosted Trees perform best in terms of absolute error (9.9 h). XGBoost Tree Ensemble performs slightly worse (11.2 h), while Multiple Linear Regression yielded a larger error (17.8 h) among the ML-based models. Polynomial Regression was excluded because its forecasts were so unstable that error metrics could not be calculated for MAE, WAPE, and R^2. This also highlights Polynomial Regression's low suitability for forecasting under the conditions of the case study.

When interpreting sMAPE, all models yield high error metrics (136-164%), with only minor differences between them. This indicates that sMAPE is highly sensitive to observations with low or zero actual values, leading to disproportionately large errors. The heavy penalties are primarily due to the inclusion of markets with low or no administrative workload (due to an absence of

customization orders for those markets) in the dataset. Markets with low or no CA orders thus increase the overall error metric for the forecast without significantly impacting the total administrative workload.

In terms of the models' ability to explain the relationships between the variables in the data, Gradient Boosted Trees achieved the highest R^2 value (0.93). R^2 can have negative values when the model performance is poor, while positive values range from 0 to 1, where strong explanatory performance is indicated by values close to 1 (Chicco et al., 2021). This implies that the Gradient Boosted Trees model captures the underlying relationships in the data slightly better than the other models, closely followed by Random Forest and XGBoost Tree Ensemble. Multiple Linear Regression also performed well, with an R^2 of 0.90.

The simple average baseline performs significantly worse across all metrics. Random Forest and Gradient Boosted Trees reduce MAE by approximately 77% compared to the Simple Average, indicating that the applied machine learning models provide a substantial improvement over a Simple Average forecast approach.

Overall, WAPE is considered the most representative error metric relative to the total administrative workload. This error metric is less sensitive to forecasting errors for markets with low or no administrative workload, which have little impact on the overall result. Instead, markets with a high volume of CA orders, and therefore requiring more administrative hours, gain greater significance. Based on WAPE, Random Forest and Gradient Boosted Trees emerge as the two ML-based models that perform best, which supports the hypothesis, as the results show that non-linear ensemble-based ML models achieve better predictive performance when compared to baseline, linear, and polynomial regression approaches.

5.5.2 Residuals

Residuals represent the difference between the actual value and the model's prediction (Actual - Prediction). They are used to assess how well the model captures the variation in the data. A residual close to zero indicates an accurate prediction. A positive value indicates an underestimation and a more negative value indicates an overestimation of the forecast. Figure 10 shows the residuals on the vertical axis and the predicted administrative workload on the horizontal axis for the Gradient Boosted Trees model. The dotted horizontal reference line represents perfect predictions.

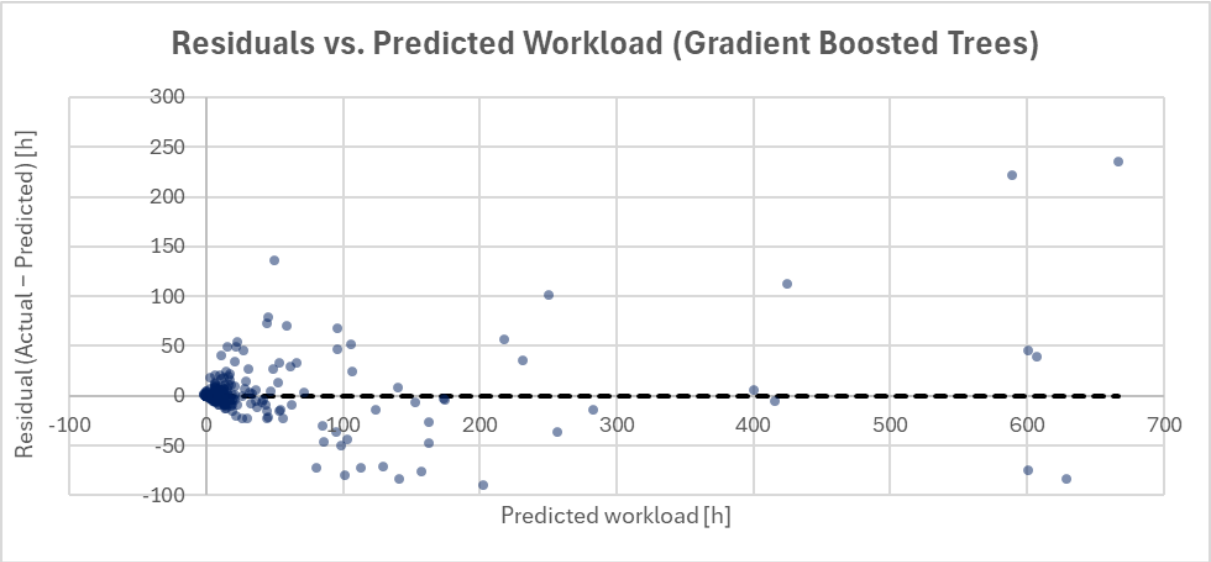


Figure 10: Residuals plotted against predicted workload for the Gradient Boosted Trees model. The horizontal line represents zero error.

The majority of the residuals are concentrated near zero in the graph, particularly at lower predicted values. This indicates that the model generally provides accurate forecasts within this range. The residuals at higher predicted workloads, conversely, show greater dispersion, suggesting that the model has more difficulty accurately predicting larger workloads. This is consistent with the data distribution, where high values occur less frequently and are therefore harder to model.

Overall, the graph shows that the model demonstrates stable predictive performance, although the increasing spread at higher levels of administrative workload indicates a reduction in accuracy for the extreme observations.

6 Discussion

This chapter discusses the findings and results in relation to the research questions and existing literature. First, it elaborates on how administrative workload is generated through interactions within the STS (RQ1). Second, it evaluates the suitability of machine learning models for forecasting administrative workload (RQ2). Third, how forecasting can support resource planning while balancing complexity is presented. Finally, sustainability in resource planning is discussed.

6.1 The generation of administrative workload

RQ 1: How is administrative workload generated within a complex, customization-intensive manufacturing environment?

Bostrom and Heinen (1977) distinguish between primary and secondary changes, in which modifications to one element of the system produce subsequent shifts in others. In the CA department, such interactions were observed across the four elements (structure, people, physical systems, and tasks).

The structural conditions in the CA department directly affect the employees. Since the volume of incoming orders is unregulated, the CA department becomes a bottleneck in the organization. This unregulated flow of orders is perceived as unpredictable by employees, making their daily workload fluctuate. Moreover, as these orders must be finished within timeframes set by the sales department, these conditions create a constant time pressure, contributing to a stressful environment. When this time pressure is sustained over longer periods, it affects stress levels and fatigue, which have direct consequences on the quality of work. These quality issues are expressed as errors in the order, but propagate to other units and later surface as rework that was never planned for. Yet, the rework must be absorbed by the CA department alongside the regular inflow of new orders. As a consequence, rework increases both the risk and frequency of ordered trucks being delivered late. This exemplifies how the interaction between structural conditions and employees' actions generates additional internal administrative workload. In line with Bostrom and Heinen's (1977) explanation of STS interactions, this can be understood as a secondary change, where conditions in one system element produce subsequent effects across other elements of the system. In addition, Anderson (1999) describes how non-linear interactions between interdependent components produce disproportionate outcomes through feedback loops. This aligns with our findings, as the inflow of new types of orders triggers interactions between the social and technical subsystems. These interactions generate a cumulative workload that exceeds what would be expected from incoming order volume alone.

The case company offers the option to order a fully customizable truck, leading to high product variation. Participant DE1 reported that although each new order follows the same process, the work content varies depending on what the customer has requested. Product variation creates a high degree of complexity in the system, which leads to higher uncertainty regarding the amount of administrative work required for each new CA order. This is consistent with ElMaraghy et al.'s (2012) observation that complexity at the product level propagates into higher levels of the system,

as the variation embedded in each order accumulates into administrative complexity across the CA department.

Product complexity has direct consequences on the CA department's daily operations. Trattner et al. (2019) found that greater customization negatively impacts lead time, consistent with our findings, in which CA orders with greater variation require more administrative hours to handle. Participant PE1 reported that preparation engineers spend considerable time on rework when design engineers make late changes to the design, as such changes require preparation engineers to review the entire design again. However, as participant DE1 noted, current design systems do not prevent design errors. As a result, design errors are prone to being passed undetected to the next unit in the workflow. To add to this, Kropsu-Vehkapera et al. (2011) describe this type of propagation, in which product complexity disrupts operational processes through delays, configuration errors, and coordination issues between units. Together, the product complexity generated by the company's customization offering creates time pressure and quality issues that are passed on through the interactions between structure, people, tasks, and physical systems.

In response to RQ1, administrative workload in complex, customization-intensive manufacturing environments results from incoming orders, as they act as the primary trigger of non-linear interactions among STS elements. Beyond this, order processing under input uncertainty within the CA department generates additional administrative work, originating from two key interactions among STS elements. First, the interaction between structure and people: unpredictable order volumes and externally imposed timelines subject employees to sustained time pressure, increasing the risk of quality errors. These errors generate rework that re-enters the system, contributing to cumulative workload through feedback loops. Second, as the complexity of the customization task increases, it demands more from the other elements in the system, intensifying the administrative effort required. Consequently, complex, customization-intensive manufacturing environments inherently generate administrative workload through internal mechanisms, where incoming order volume does not correspond proportionally to the administrative effort required, and this workload is further amplified by non-linear interactions among STS elements.

6.2 The suitability of machine learning for forecasting administrative workload

RQ 2: How does the predictive performance of different forecasting methods compare when applied to administrative workload data in a complex, customization-intensive manufacturing environment?

The results demonstrate that non-linear ensemble-based ML models achieve stronger predictive performance than Simple Average, linear, and polynomial approaches for administrative workload forecasting in complex, customization-intensive manufacturing environments, thereby answering RQ2. Random Forest and Gradient Boosted Trees, both ensemble-based methods, achieved the lowest MAE and WAPE, whereas Multiple Linear Regression showed weaker performance. This can be explained by the algorithmic structure of ensemble-based models. While Multiple Linear Regression estimates one global linear relationship between the input variables and administrative workload, ensemble-based models combine multiple learners to improve predictive accuracy (Alam et al., 2025). More specifically, Random Forest and Gradient Boosted Trees learn local prediction rules by recursively splitting the data into smaller regions. This allows them to capture non-linear

relationships and interaction patterns between variables such as market, period, and number of CA jobs. The performance metrics in the study indicate that models capable of identifying more complex, non-linear patterns are needed, as the underlying relationships in the data are not sufficiently captured by the simpler models. This aligns with the explanation that ML-based models excel at describing patterns in data that predefined mathematical equations struggle to capture (Alam et al., 2025; Petropoulos et al., 2022). Furthermore, the Simple Average baseline performs significantly worse across all error metrics, providing a useful comparison of the models' performance against an easily understandable benchmark, which should be done in order to assess the model's performance (Hewamalage et al., 2023; Petropoulos et al., 2022). In both this study and De Simone et al. (2025), non-linear ensemble-based ML models outperform both simple baseline, linear, and polynomial models when forecasting workload in manufacturing environments.

However, the comparison between the models is influenced by the selection of error metrics. While WAPE distinguishes the best-performing models from low-performers, sMAPE displays high error metrics for all models, which makes them hard to differentiate. This pattern can be explained by the structure of the dataset, as it includes a large number of observations with low or zero workload. sMAPE disproportionately penalizes low or zero observations as it is sensitive when both the actual value and predicted value are small and have a slight difference (a denominator close to zero). WAPE provides a more representative performance measure in this study as it relates forecast error to the overall workload. WAPE ensures that prediction errors on markets related to high workload influence the error metric more than markets related to low workload, which provides a better reflection of the case company's business. This highlights the importance of selecting error metrics that reflect the practical purpose of the forecast and the characteristics of the data (Hewamalage et al., 2023).

In this study, the key difference from previous studies is the variation and size of the dataset. De Simone et al. (2025, p. 16) stated: "A larger and more diverse dataset would not only facilitate better model training but also improve the statistical power of the evaluation [...]". The dataset used in this study is 15 times larger, enabling a more robust model training and strengthening the reliability and generalizability of the results. Overall, the results demonstrate how non-linear ensemble-based ML forecasts can be trained and used in complex, customization-intensive manufacturing environments and perform better than Simple Average, linear, and polynomial approaches.

6.3 Improving resource planning through an integrated understanding of administrative work and forecasting

Integrating sociotechnical considerations into the forecasting process is a key contribution. Unlike prior studies that focused primarily on technical model performance (e.g., De Simone et al., 2025), this study takes into account how administrative work is created and performed within the case organization. Thus, forecasting models can be grounded in the operational reality of the case environment, where ML is combined with an understanding of system complexity and administrative work practices. This aligns with the notion that successful implementations of ML

for decision-making depend on model performance, the quality, and contextual relevance of the data the model uses for training (Sarker, 2021).

The complexity of the CA department poses a challenge to the case company's ability to plan its resources. Reducing or eliminating this complexity is not necessarily the ultimate goal. ElMaraghy et al. (2012, p. 811) state that "the ultimate goal is not only to mitigate potential negative effects of complexity but also to maximize the benefits from it, such as using it to gain a competitive edge in the market". For the case company, this means continuing to deliver competitive, customer-adapted trucks while simultaneously balancing complexity within the CA department. As the proportion of customization orders continues to increase for the case company, it is necessary to adapt to a complex operating environment and regard complexity as a persistent condition rather than a temporary challenge. A higher degree of complexity in the system calls for the ability to plan resources for a future in which complexity is managed rather than simply reduced. Thus, to handle future customization demands, administrative workload forecasting is needed to support the mitigation of the negative effects of complexity.

Complexity stems from new customer CA order requests and the activities required to handle them. The volume and variability in new orders make the administrative workload difficult to oversee and the entire system complex. Petropoulos et al. (2022) highlight the emergence of new products as a challenge for forecasting demand. The continuous demand for new truck customizations creates a similar challenge, as uncertainty about customers' preferences complicates administrative work planning, highlighting the need for tools such as forecasting to support resource planning under uncertainty. In addition, the constant inflow of new types of customization orders forces employees to develop new solutions, which creates work they perceive as more complex. ML-based forecasts can capture complex relationships among variables relevant to workload, but doing so requires a strong understanding of the forecast context (Kara & Şengüler, 2025; Sepaspour et al., 2025). The ML-based models in the study capture relationships among variables, especially in markets with low administrative workload.

However, the residuals indicate that peaks of high administrative workload are difficult for the ML-based models to predict, as these peaks deviate from historical patterns and therefore occur outside the relationships learned during model training. This suggests that while ML-based models are useful for capturing general patterns, they are less reliable in handling extreme or irregular events. In such cases, organizations must have the resilience to adapt to varying levels of workload. New approaches to resource planning may be needed, where capacity is planned based on forecast insights and resource buffers informed by managerial judgement to account for unpredictable events. The peaks in the study could be explained by unexpected late-arriving orders or rework due to prior administrative errors. The peaks were primarily observed in markets with the most advanced CA orders, consistent with Trattner et al.'s (2019) claim that high product complexity negatively impacts manufacturing costs, lead time, processing time, and process productivity. Overall, departmental complexity has both advantages (competitive edge) and disadvantages (impact on time, cost, and productivity). The benefits and drawbacks must therefore be balanced to maximize overall benefit, and forecasting in resource planning can support this balancing.

Finally, the use of an MMR design was intended to strengthen the credibility, validity, and reliability of the study's conclusions. The MMR design involves trade-offs, primarily in the difficulty of executing the study (Johnson & Onwuegbuzie, 2004). Advantageously, it combines the strengths and minimizes the weaknesses of each research paradigm. In this study, the qualitative phase provided contextual understanding of how administrative workload is generated, while the quantitative phase evaluated the predictive performance of forecasting models. This enabled the study to address a more complex resource planning problem than would have been possible through a purely qualitative or quantitative approach alone. By being able to address a wider range of research questions (Johnson & Onwuegbuzie, 2004), the MMR design enables a broader and more generalizable contribution.

6.4 Sustainability in resource planning

Beyond operational improvements, the results also highlight implications for sustainability in complex, customization-intensive manufacturing environments. The lack of informed resource planning in the CA department not only results in operational problems but also creates unsustainable conditions within the CA department. As demonstrated in the qualitative findings, fixed delivery deadlines and an unpredictable inflow of orders from the sales department create a persistent time pressure. This, in turn, leads to errors, rework, and work intensification.

From a sustainability perspective, inadequate resource planning in complex, customization-intensive manufacturing environments contributes to internal unsustainability, creating rippling effects in social, economic, and environmental dimensions. Socially, the fatigue from persistent time pressure and work intensification affects employee performance and well-being, which aligns with Mariappanadar's (2014) argument that sustained work intensification poses a health risk and negatively affects employees' cognitive abilities. Economically, the rework, inefficiencies, and delays generated within the CA department represent the economic consequences of the misallocation of resources. Problems in the CA department are solved temporarily, and little attention is paid to long-term resource decisions, as they are difficult to make without support from future forecasts. Environmentally, short-term thinking can lead to less effective resource planning with reactive material allocation, which limits the optimization of goods in transport. Along with an increased need for rushed transports with low fill rates, this can result in a higher environmental impact from transportation. Consequently, inadequate resource planning hinders the development of long-term stability and sustainability within the CA department. This supports Bansal and DesJardine's (2014) view that managers tend to prioritize short-term needs over investments that support long-term sustainability.

In the study, forecasting does not simply improve accuracy in resource planning but also contributes to mitigating the internal generation of unsustainable workload. When relating these findings to the Sustainable Development Goals (SDGs), improved resource planning through forecasting can contribute to both SDG 8 (Decent Work and Economic Growth) and SDG 9 (Industry, Innovation, and Infrastructure). Reduced work intensification and improved working conditions align with SDG 8. Advancements in industrial processes and innovation are reflected through the use of ML-based forecasting to support resource planning in complex, customization-

intensive manufacturing environments, in line with SDG 9. However, improvements in one dimension may cause conflicts with others as the SDGs generate both synergies and trade-offs, as highlighted by Pradhan et al. (2017). In this study, improvements in resource planning through forecasting appear to create more immediate synergies between social and economic sustainability, while environmental sustainability effects may become visible over a longer time horizon.

7 Conclusion

This chapter addresses the purpose of this study. Additionally, theoretical contributions, practical implications, limitations, and future research are presented.

The purpose of this study was to investigate how resource planning can be improved in complex, customization-intensive manufacturing environments by building a better understanding of administrative work practices, integrated with the application of ML-based forecasting models. Together, the two research questions demonstrate that improvement requires combining contextual understanding with technical forecasting tools. Although non-linear ensemble-based ML forecasting models show limitations in predicting extreme or irregular events, particularly those driven by high complexity and customization, they can be complemented with an understanding of underlying work practices to be effectively used for resource planning. Instead of simply reducing system complexity, it can be balanced. In this way, forecasting can improve resource planning by enabling organizations to navigate the trade-offs of complexity inherent in complex, customization-intensive manufacturing environments.

7.1 Theoretical contributions

This study contributes to the operations management literature by extending the conceptualization of administrative workload in complex, customization-intensive manufacturing environments. Previous research has primarily focused on workload in shop-floor settings and the performance of forecasting models (e.g., De Simone et al., 2025), overlooking administrative work and the sociotechnical conditions in which workload emerges. Thus, by applying ML-based models to administrative work in complex, customization-intensive manufacturing environments and demonstrating how ensemble-based models achieve stronger performance compared to simpler models, the study extends forecasting research. Drawing on STS literature, which conceptualizes work systems as interactions between social and technical elements (Bostrom & Heinen, 1977; Kemp et al., 2024), this study reframes administrative workload as both generated by incoming orders and internally amplified through interactions between structure, people, tasks, and physical systems in the STS. Furthermore, the study shows how the STS perspective can be operationalized within a forecasting process. This integration was achieved through an exploratory sequential MMR design, where qualitative findings inform a subsequent quantitative phase (Schoonenboom & Johnson, 2017). In doing so, the study provides a novel application of MMR design within the field of industrial engineering and management by demonstrating how administrative work practices and technical forecasting evaluation can be integrated in the study of resource planning in complex, customization-intensive manufacturing environments. Instead of treating forecasting as a purely technical model-selection problem, the research design shows how contextual understanding can shape the formulation, development, and interpretation of forecasting models. The mapping of the CA department's administrative work in the qualitative study provided this contextual understanding. Specifically, it led to insights into the relevant data, the target variable, the aggregation level of data, the input variables, and the subsequent interpretation of model results, which informed the subsequent CRISP-DM phases (Data Preparation, Modeling, and Evaluation) in the quantitative study. In this way, the study demonstrates that forecasting administrative

workload for resource planning in complex, customization-intensive manufacturing environments is a context-dependent process in which administrative work practices shape the conditions for forecasting and interpretation of the results.

7.2 Practical implications

Practical implications have been developed based on the case study for complex, customization-intensive environments. The recommendations are intended for managers responsible for resource planning in organizations facing the same challenges as the case company in the study. Specifically, they are aimed at managers who lack structured methods for gaining insights into future administrative workloads, even though historical workload data is available. The consequences of this are reactive planning, persistent time pressure, and difficulties in allocating resources across administrative functions. Although the case study was conducted in a manufacturing environment, we argue that the practical application is also suitable for other complex environments that require a high degree of customization and possess usable historical data. For instance, resource planning in healthcare, where medical departments handle complexity and specific customized services for patients.

The recommendations are structured as a five-step guide in Figure 11 and show how managers can develop an understanding of the administrative workload, use these insights to develop forecasts, and use them in resource planning. The guide is based on the execution of our study and is described in the steps outlined. First, begin with mapping the administrative work system to understand how administrative workload is generated and what the aim of a forecasting tool would be in resource planning. Second, translate the qualitative understanding into data requirements by defining the target variable and selecting input variables that are also available at the time of forecasting. Ensure that data for historical administrative work is available or can begin to be collected, and decide the appropriate forecast scope. Third, select a data analytics tool and prepare the data by extracting workload and other relevant data from internal systems, aggregating it to the chosen forecast scope, cleaning the data, and merging datasets when required. Clean duplicates, missing values, and inconsistencies before modeling. Fourth, train ML models and develop a simple baseline. Use a time-based train/test split to reflect real forecasting and assess performance with context-relevant metrics and residual analysis to identify where the models perform well or poorly. Finally, apply prepared input data to the selected models and use the forecast output in resource planning by integrating it into planning dashboards. Combine the forecast with managerial judgement to improve resilience against unpredictable workloads, especially since the models may struggle to capture extreme workload outliers, as in this study. This enables managers to reallocate staff, adjust priorities, prepare overtime, or escalate bottlenecks to support more proactive resource planning. The forecast output for resource planning was applied in a pilot project at the case company, providing practical support for the recommendation. Each step of the recommendations is described in more detail in Figure 11.

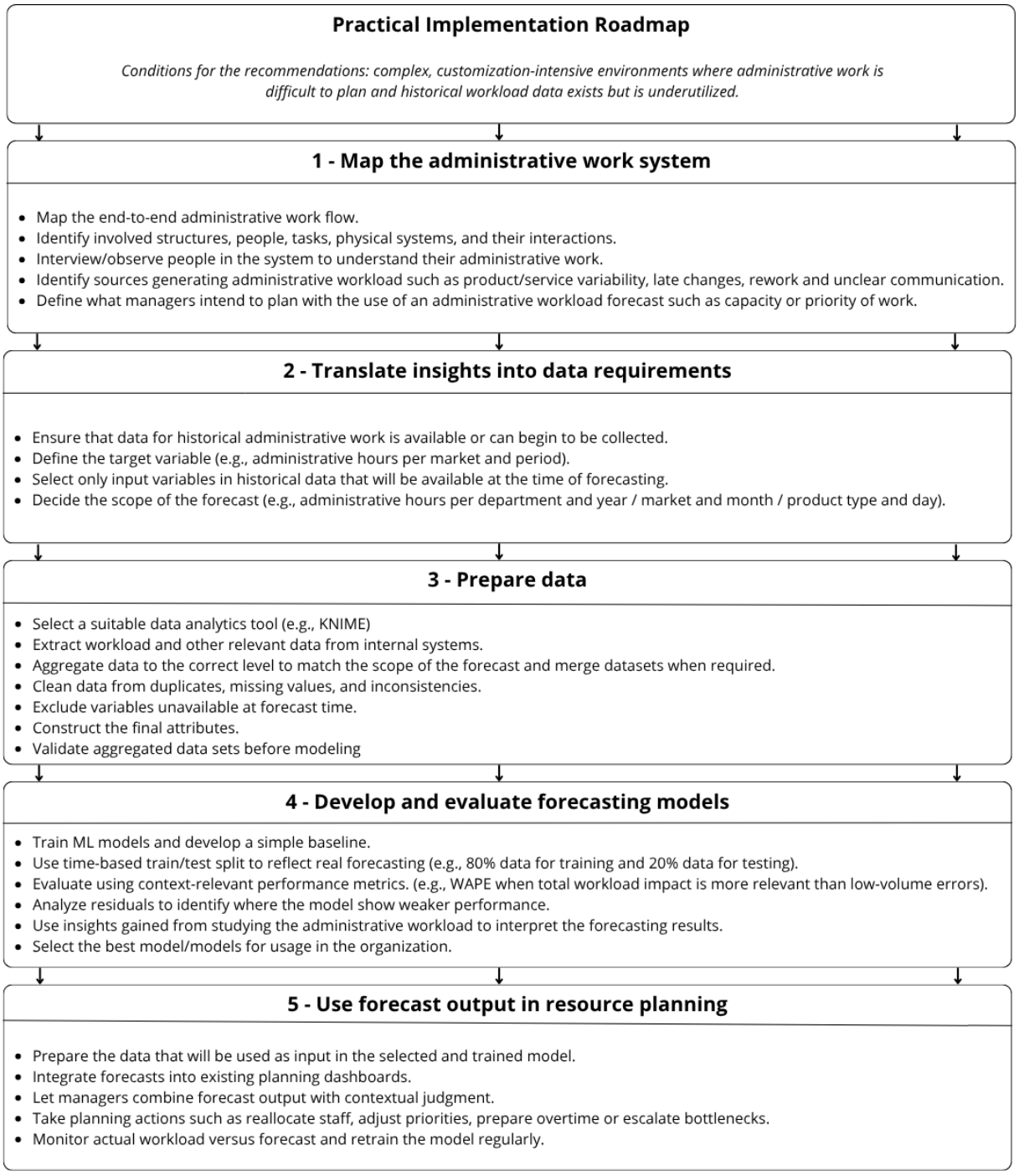


Figure 11: Practical implementation framework for administrative workload forecasting in complex, customization-intensive environments where historical workload data exists but is underutilized.

7.3 Limitations and future research

This study is subject to several limitations. First, the study focuses on resource planning over a relatively short timeframe, and the test period covered five months. This limits the ability to assess how stable the models' predictive performance would be across longer time horizons with seasonal patterns. Second, the target variable is based on administrative hours data, making the forecast dependent on the data's ability to describe the CA department's conditions. In addition, the selection of explanatory variables was constrained by data availability. Third, it is important to

emphasize that the specific models that demonstrated the best performance in this study may not perform equally in a different context, even if the suggested process in the study is generalizable. Therefore, the performance of different models must always be tested if applied in new contexts. Finally, the study places greater emphasis on the early phases (from Business Understanding to the Evaluation phase) of the CRISP-DM framework, while less attention is given to the Deployment phase and organization scaling of the ML-based forecast. As a result, the long-term effects of resource planning using the proposed forecasting approach cannot be assessed.

Based on the study's limitations, future research could address these through new studies. First, extending the time horizon of collected training data may improve the ability of ML-based models to detect seasonal patterns in data and other long-term effects. Second, studies should also examine how complex, customization-intensive manufacturing environments can improve their data, as forecasting performance depends on how well the data reflects actual processes. Furthermore, segmentation approaches, where markets or types of customization orders are modeled separately, could be explored to improve predictive performance under different conditions. Finally, future research could build directly on this study by examining how ML-based administrative workload forecasts are translated into resource planning practices after implementation. While this thesis demonstrates how administrative workload can be conceptualized and forecasted, further research is needed to understand how forecast outputs influence managerial decision-making, employee workload allocation, prioritization, and the balancing of complexity in daily operations. Such studies could explore how to apply forecasting insights to resource planning and extend the study's practical implications.

References

Alam, M. N., Laxmi, V., Sharma, A., & Dangi, S. (2025). Machine Learning: Key Algorithms, Practical Applications, and Current Research Directions. *International Journal of Electrical and Electronics Engineering*, *12*(4), 12-46.

Anderson, P. (1999). Complexity Theory and Organization Science. *Organization Science (Providence, R.I.)*, *10*(3), 216–232. https://doi.org/10.1287/orsc.10.3.216

Armstrong, J.S. (2001). *Principles of Forecasting*. Springer US. https://doi.org/10.1007/978-0-306-47630-3

Bansal, P., & DesJardine, M. (2014). Business sustainability: It is about time. *Strategic Organization, 12*(1), 70–78. https://doi.org/10.1177/1476127013520265

Bell, E., Bryman, A. & Harley, B. (2022). *Business research methods* (6 ed.). Oxford University Press.

Bostrom, R. P., & Heinen, J. S. (1977). MIS Problems and Failures: A Socio-Technical Perspective. Part I: The Causes. *MIS Quarterly*, *1*(3), 17–32.

Chapman, P., Clinton, J., Kerber, R., Khabaza, T., Reinartz, T., Shearer, C., & Wirth, R. (2000). *CRISP-DM 1.0: Step-by-step data mining guide* (CRISP-DM Consortium). https://www.semanticscholar.org/paper/CRISP-DM-1.0%3A-Step-by-step-data-mining-guide-Chapman/54bad20bbc7938991bf34f86dde0babfbd2d5a72

Chicco, D., Warrens, M. J., & Jurman, G. (2021). The coefficient of determination R-squared is more informative than SMAPE, MAE, MAPE, MSE and RMSE in regression analysis evaluation. *PeerJ. Computer Science, 7*, Article e623. https://doi.org/10.7717/peerj-cs.623

Da Silveira, G., Borenstein, D., & Fogliatto, F. S. (2001). Mass customization: Literature review and research directions. *International Journal of Production Economics, 72*(1), 1–13. https://doi.org/10.1016/S0925-5273(00)00079-7

De Simone, V., Di Pasquale, V., Calabrese, J., Miranda, S., & Iannone, R. (2025). A Supervised Machine Learning-Based Approach for Task Workload Prediction in Manufacturing: A Case Study Application. *Machines (Basel), 13*(7), 602. https://doi.org/10.3390/machines13070602

Dubois, A., & Gadde, L.-E. (2002). Systematic combining: an abductive approach to case research. *Journal of Business Research, 55*(7), 553–560. https://doi.org/10.1016/S0148-2963(00)00195-8

Eisenhardt, K. M. (1989). Building Theories from Case Study Research. *The Academy of Management Review*, *14*(4), 532. https://doi.org/10.2307/258557

ElMaraghy, W., ElMaraghy, H., Tomiyama, T., & Monostori, L. (2012). Complexity in engineering design and manufacturing. *CIRP Annals*, *61*(2), 793–814. https://doi.org/10.1016/j.cirp.2012.05.001

Fife, S. T., & Gossner, J. D. (2024). Deductive Qualitative Analysis: Evaluating, Expanding, and Refining Theory. *International Journal of Qualitative Methods, 23*. https://doi.org/10.1177/16094069241244856

Fogliatto, F. S., da Silveira, G. J. C., & Borenstein, D. (2012). The mass customization decade: An updated review of the literature. *International Journal of Production Economics, 138*(1), 14–25. https://doi.org/10.1016/j.ijpe.2012.03.002

Gatta, R., Vallati, M., Pirola, I., Lenkowicz, J., Tagliaferri, L., Cappelli, C., & Castellano, M. (2020). An empirical analysis of predictors for workload estimation in healthcare. In V. V. Krzhizhanovskaya, G. Závodszky, M. H. Lees, J. J. Dongarra, P. M. A. Sloot, S. Brissos, & J. Teixeira (Eds.), *Computational Science – ICCS 2020* (Vol. 12137, pp. 304–311). Springer. https://doi.org/10.1007/978-3-030-50371-0

Hallén, L., Johanson, J., & Seyed-Mohamed, N. (1991). Interfirm Adaptation in Business Relationships. *Journal of Marketing*, *55*(2), 29–37.

Han, J., Kamber, M., & Pei, J. (2012). *Data mining: concepts and techniques, third edition* (3rd ed.). Morgan Kaufmann Publishers.

Hemlata, C., & Gulia, P. (2019). Experimental evaluation of open source data mining tools: R, rapid miner and knime. *International Journal of Innovative Technology and Exploring Engineering, 9*(1).

Hewamalage, H., Ackermann, K., & Bergmeir, C. (2023). Forecast evaluation for data scientists: common pitfalls and best practices. *Data Mining and Knowledge Discovery, 37*(2), 788–832. https://doi.org/10.1007/s10618-022-00894-5

Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: principles and practice*, 3rd edition, OTexts. https://otexts.com/fpp3/. Retrieved: 2026-02-05.

Hyndman, R. J., & Koehler, A. B. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting*, *22*(4), 679–688. https://doi.org/10.1016/j.ijforecast.2006.03.001

Johnson, R. B. (2017). Dialectical Pluralism: A Metaparadigm Whose Time Has Come. *Journal of Mixed Methods Research, 11*(2), 156–173. https://doi.org/10.1177/1558689815607692

Johnson, R. B., & Onwuegbuzie, A. J. (2004). Mixed Methods Research: A Research Paradigm Whose Time Has Come. *Educational Researcher, 33*(7), 14–26. https://doi.org/10.3102/0013189X033007014

Johnson, R. B., & Schoonenboom, J. (2016). Adding Qualitative and Mixed Methods Research to Health Intervention Studies: Interacting With Differences. *Qualitative Health Research*, *26*(5), 587–602. https://doi.org/10.1177/1049732315617479

Kara, B., & Şengüler, H. (2025). A Comparative Analysis of Budget Forecasting Methods: A Systematic Literature Review Covering the 1983–2024 Period. *Public Budgeting & Finance*. 1-18. https://doi.org/10.1111/pbaf.70008

Kemp, T., Ayton, J., Butler-Henderson, K., & Lam, M. (2024). Using socio-technical systems theory to study the health information management workforce in Australian acute hospitals. *Social Theory & Health*, *22*(4), 285–300. https://doi.org/10.1057/s41285-024-00214-5

Kropsu-Vehkapera, H., Haapasalo, H., Lokkinen, S., & Phusavat, K. (2011). The influence of product complexity on order handling process. *International Journal of Innovation and Learning*, *10*(2), 123–143. https://doi.org/10.1504/IJIL.2011.041793

Lee, H. L. (2002). Aligning Supply Chain Strategies with Product Uncertainties. *California Management Review, 44*(3), 105–119. https://doi.org/10.2307/41166135

Mariappanadar, S. (2014). Stakeholder harm index: A framework to review work intensification from the critical HRM perspective. *Human Resource Management Review*, *24*(4), 313–329. https://doi.org/10.1016/j.hrmr.2014.03.009

Mulcahy, L., Rossner, M., & Tsalapatanis, A. (2021). It's about time: investigating the temporal in socio-legal studies through unstructured interviews. *Journal of Law and Society*, *48*(S1), S104–S117. https://doi.org/10.1111/jols.12330

Onwuegbuzie, A. J., & Johnson, R. B. (2006). The Validity Issue in Mixed Research. *Research in the Schools*, *13*(1), 48.

Petropoulos, F., Apiletti, D., Assimakopoulos, V., Babai, M. Z., Barrow, D. K., Ben Taieb, S., Bergmeir, C., Bessa, R. J., Bijak, J., Boylan, J. E., Browell, J., Carnevale, C., Castle, J. L., Cirillo, P., Clements, M. P., Cordeiro, C., Cyrino Oliveira, F. L., De Baets, S., Dokumentov, A., … Ziel, F. (2022). Forecasting: theory and practice. *International Journal of Forecasting, 38*(3), 705–871. https://doi.org/10.1016/j.ijforecast.2021.11.001

Pradhan, P., Costa, L., Rybski, D., Lucht, W., & Kropp, J. P. (2017). A systematic study of sustainable development goal (SDG) interactions. *Earth's future, 5*(11), 1169-1179. https://doi.org/10.1002/2017EF000632

Qu, S. Q., & Dumay, J. (2011). The qualitative research interview. Qualitative *Research in Accounting and Management, 8*(3), 238–264. https://doi.org/10.1108/11766091111162070

Sarker, I. H. (2021). Machine Learning: Algorithms, Real-World Applications and Research Directions. *SN Computer Science*, *2*(3), 160. https://doi.org/10.1007/s42979-021-00592-x

Schoonenboom, J., & Johnson, R. B. (2017). How to Construct a Mixed Methods Research Design. *Kölner Zeitschrift Für Soziologie Und Sozialpsychologie*, *69*(Suppl 2), 107–131. https://doi.org/10.1007/s11577-017-0454-1

Schröer, C., Kruse, F., & Gómez, J. M. (2021). A Systematic Literature Review on Applying CRISP-DM Process Model. *Procedia Computer Science, 181*, 526–534. https://doi.org/10.1016/j.procs.2021.01.199

Sepaspour, R., Hajikarimi, P., & Moghadas Nejad, F. (2025). Future climate prediction and projection: A systematic review of classical and advanced methodologies. *Theoretical and Applied Climatology, 156*(10), Article 527. https://doi.org/10.1007/s00704-025-05795-3

Trattner, A., Hvam, L., Forza, C., & Herbert-Hansen, Z. N. L. (2019). Product complexity and operational performance: A systematic literature review. *CIRP Journal of Manufacturing Science and Technology*, *25*, 69–83. https://doi.org/10.1016/j.cirpj.2019.02.001

Yin, R.K. (2018). *Case study research and applications: design and methods*. (Sixth edition). SAGE.